

AK High-Dimensional Projection Structural Theory

Version 18.0.0: Collapse Structures, Group Simplification, and Persistent Projection Geometry

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1 Chapter 1: Scope, Claim Taxonomy, Guard-Rails, and Minimal Manifest Axiom

Scope Box (Core guard-rails; repaired after-collapse and audit-ready). All statements in this chapter, and all subsequent statements labeled as *Core*, are made under the following guard-rails.

- **Constructible one-parameter persistence over a field k .** We work in constructible one-parameter persistence over a field k :

$$\text{Pers}_k^{\text{cons}} \subset [(\mathbb{R}, \leq), \text{Vect}_k].$$

Filtered (co)limits are computed objectwise in $[\mathbb{R}, \text{Vect}_k]$ and used only after verification of constructibility, or after passage to declared windowed collapsed persistence profiles.

- **Barcode-level positive-threshold collapse.** For $\tau > 0$, the Core uses the barcode-level threshold deletion operation

$$T_\tau^{\text{bar}}$$

defined in Chapter 2 and Appendix A. In the general Core regime, T_τ^{bar} is endpoint-independent, idempotent, threshold-monotone, and non-expansive at the barcode metric level. It is not asserted to be a global exact Serre reflector on all of $\text{Pers}_k^{\text{cons}}$.

- **Windowed evaluation first.** Windows are right-open intervals $W = [u, u')$. For a filtered object F , monitored degree i , window W , and threshold τ , the Core evaluates the collapsed persistence profile by the fixed order

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is part of the Core.

- **Quantale-enriched audit geometry; definable right-open windows.** The time index $(\mathbb{R}, \leq)$ is equipped with a commutative quantale

$$(\mathbb{V}, \otimes, \mathbb{I}, \leq)$$

for distances and budgets. Whenever global gluing is invoked, the relevant window cover is required to be definable (o-minimal or Denef-Pas, as appropriate) and to have finite Čech depth.

- **Representative discipline.** A filtered representative

$$C_{\tau, W}F$$

is not a global collapse functor by default. It is an admissible representative of the windowed collapsed persistence profile

$$\text{Eval}_{i, W, \tau}(F)$$

in the monitored degrees. Any derived or Ext-level use of $C_{\tau, W}F$ requires a realization-compatible representative and the eligible realization regime.

- **Bridge policy.** The forward bridge

$$\text{PH}_1 \Rightarrow \text{Ext}^1$$

is part of the Core only under the eligible realization regime of Chapter 3. Any reverse use

$$\text{Ext}^1 \Rightarrow \text{PH}_1$$

is never part of the global Core and may be used only under separately declared local safety conditions in the Interface/Search layer. No unconditional global equivalence

$$\text{PH}_1 \Leftrightarrow \text{Ext}^1$$

is claimed.

- **Tower comparison discipline.** Type IV diagnostics require a declared tower comparison artifact

$$\phi_{i, W, \tau} : \text{ColimProfile}_{i, W, \tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i, W, \tau}(F_\infty).$$

Such a comparison is automatic only in a separately declared exact-admissible regime.

- **Audit-first doctrine.** The Core is governed by the budget inequality

$$\Sigma \delta < \text{gap}_\tau.$$

Search quantities, including any scalar potential Φ , may guide proposal generation, but do not alter Core acceptance criteria.

- **Manifest discipline.** Every certified proof object must carry declared windows, threshold policy, collapse policy, representative policy if representatives are used, ledger policy, diagnostics, gate verdicts, tower comparison artifacts if Type IV diagnostics are invoked, and artifact references. The detailed serialization format may live outside the main text, but the existence of these data is a Core audit requirement.

Accordingly, this chapter fixes the admissible language for the whole text: what counts as a theorem, what counts as an operational policy, what counts as an interface-level statement, what counts only as a search-side artifact, and what is explicitly excluded from the claimed Core.

1.0. Claim taxonomy and document discipline

Definition 1.1 (Claim Taxonomy). Every mathematically meaningful statement in this document belongs to one of the following six classes.

- (i) **Core theorem.** A statement proved inside the standing Core regime: constructible one-parameter persistence over a field, repaired windowed collapse evaluation, declared diagnostics, and the fixed eligible realization discipline where Ext is invoked. Core theorems may be used freely in later Core arguments, subject to their hypotheses.
- (ii) **Core lemma.** A local or supporting statement proved in the same regime as a Core theorem, but used primarily as an internal step in later Core arguments.
- (iii) **Operational policy.** A declared protocol governing how certification is performed: for example, windowing rules, threshold policy, collapse policy, representative policy, budget checks, overlap checks, diagnostic conventions, or manifest requirements. Operational policies are binding on admissible runs, but are not themselves theorems about external mathematical objects.
- (iv) **Interface theorem.** A theorem of the form

$$(\text{declared problem-specific input}) \implies (\text{Core-level obstruction or discharge}).$$

Such a theorem may use the Core, but it does not enlarge the Core. It is valid only relative to the declared realization, transport assumptions, and interface hypotheses.

- (v) **Search artifact.** Any output of proposal generation, navigation, terrain mapping, heuristic prioritization, or AI-assisted exploration: for example, a validity map, a candidate list, a search trajectory, a scalar potential landscape, or an agent-generated report. A Search artifact is never, by itself, a proof object.
- (vi) **Non-claim.** A statement explicitly excluded from the asserted content of the document. Typical examples include unconditional global reverse bridges, positive-threshold exact reflector claims, problem-resolution claims without declared interface hypotheses, or search outputs promoted directly to theorems.

Remark 1.2 (Interpretive rule). The role of the taxonomy is not cosmetic. It enforces a separation of mathematical authority: Core theorems govern proof, Operational policies govern admissible certification, Interface theorems govern conditional transport to specific domains, and Search artifacts govern only proposal generation. Whenever the status of a statement is ambiguous, the weaker classification prevails.

1.1. Repaired Unified Collapse Contract

Theorem 1.3 (Repaired Unified Collapse Contract: mathematical core). *Fix a commutative quantale $\mathbb{V}$ for distances and budgets, and fix right-open windows. In the constructible one-parameter regime, the following repaired Core package is available.*

- (i) **Barcode-level threshold collapse.** For each $\tau > 0$, the operation

$$T_{\tau}^{\text{bar}}$$

deletes finite bars of length at most τ . It is endpoint-independent, idempotent, threshold-monotone, and non-expansive at the barcode metric level.

- (ii) **Windowed evaluation.** For every monitored degree i , window W , and threshold τ , the Core evaluates

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

This is the default after-collapse persistence profile used by gate checks.

(iii) **No unconditional crop–collapse commutation.** The identity

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is not asserted in general. It may be used only under a declared τ -window-adapted condition or inside an exact-admissible regime.

(iv) **Representative compatibility.** If a filtered representative $C_{\tau, W}F$ is used, it must satisfy

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F)$$

for all monitored degrees i , under the declared representative equivalence policy.

(v) **Deletion-type stability.** After applying the repaired windowed evaluation, deletion-type updates are monotone at the retained-profile level, while inclusion-type updates are stability-only and non-expansive in the declared metric sense.

(vi) **Exact-admissible strengthening.** Stronger claims such as strict functoriality, exactness, or canonical tower comparison after positive-threshold deletion may be used only in a separately declared τ -split or exact-admissible regime.

Policy 1 (Repaired Unified Collapse Contract: audit policy). A certified run governed by the repaired UCC must declare:

(i) the window family;

(ii) the threshold policy;

(iii) the collapse policy:

$$\text{barcode_level}, \quad \text{tau_split}, \quad \text{exact_admissible};$$

(iv) the evaluation order, normally

$$\mathbf{P}_i \rightarrow W_W \mathbf{P}_i \rightarrow T_\tau^{\text{bar}}(W_W \mathbf{P}_i);$$

(v) the representative policy, if $C_{\tau, W}F$ is used;

(vi) the δ -ledger aggregation law;

(vii) the gate protocol;

(viii) the safety margin gap_τ ;

(ix) the budget comparison

$$\Sigma \delta < \text{gap}_\tau;$$

(x) tower comparison artifacts, if Type IV diagnostics are invoked.

The δ -ledger is additive in the declared budget semantics, subadditive under composition, and non-increasing under after-collapse 1-Lipschitz post-processing at the monitored profile level.

Proof and status. Theorem 1.3 records the repaired mathematical Core package. Policy 1 records the operational audit discipline. The two are intentionally separated: the former is theorem-bearing, while the latter governs admissible certification.

1.2. UCC-Contract, windows, and the repaired after-collapse evaluation rule

Definition 1.4 (UCC-Contract). A *UCC-Contract* on a certified run consists of the following data.

- (1) A quantale $\mathbb{V}$ governing distances and budgets.
- (2) A right-open MECE family of windows $W = [u, u')$.
- (3) A threshold policy τ , or a declared τ -sweep.
- (4) The barcode-level collapse operation T_τ^{bar} .
- (5) The repaired evaluation rule

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

- (6) A representative policy for $C_{\tau,W}F$, if filtered or derived-level arguments are used.
- (7) A δ -ledger policy with admissible aggregation law.
- (8) A gate protocol specifying which diagnostics are checked and in which order.
- (9) A tower comparison policy, if Type IV diagnostics are invoked.

Definition 1.5 (Windows and MECE discipline). A *domain window* is a right-open interval $W = [u, u') \subset \mathbb{R}$. A family $\{W_\alpha\}$ is *MECE* if it is mutually exclusive and collectively exhaustive on the declared range. Whenever local certificates are glued, the cover must be accompanied by a declared overlap structure and, when relevant, a finite Čech-depth bound.

Definition 1.6 (Window restriction). For a window $W = [u, u')$, define

$$W_W : \text{Pers}_k^{\text{cons}} \rightarrow \text{Pers}_k^{\text{cons}}$$

by restriction to W , with extension by zero outside W when a global-indexed object is required. At the barcode level, this operation clips interval summands to the window.

Definition 1.7 (Windowed collapsed profile). For $F \in \text{FiltCh}(\cdot)k$, monitored degree i , window W , and threshold $\tau > 0$, define

$$\mathcal{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Definition 1.8 (Pipeline budget). For per-step budget contributions $\delta_j \in \mathbb{V}$, the total budget on a monitored degree i is

$$\Sigma\delta(i, \{\tau_j\}) := \bigoplus_j \delta_j(i, \tau_j).$$

The quantity $\Sigma\delta$ is a certification datum. A scalar potential Φ may be derived from $\Sigma\delta$ in a separate Search layer, but Φ is not part of the Core acceptance predicate.

Remark 1.9 (Core sovereignty over Search). The Core accepts or rejects only through declared gate conditions, diagnostics, budget checks, and manifest-supported artifacts. No Search artifact — including a terrain map, a proposed descent direction, a heuristic score, a validity map, or an AI confidence value — upgrades a failed Core verdict to a passing one. This asymmetry is structural and not merely stylistic.

1.3. Overlap Gate and B-Gate⁺

Definition 1.10 (Overlap Gate). Let W_α and W_β be overlapping windows. The *Overlap Gate* passes if the following hold on the overlap $W_\alpha \cap W_\beta$.

- (i) The overlap collapsed profiles agree up to the declared overlap budget:

$$d_{\text{int}}(\text{Eval}_{i, W_\alpha \cap W_\beta, \tau}(F_\alpha), \text{Eval}_{i, W_\alpha \cap W_\beta, \tau}(F_\beta)) \leq \Sigma \delta_{\alpha\beta}.$$

- (ii) The safety margin dominates the overlap budget:

$$\text{gap}_\tau > \Sigma \delta_{\alpha\beta}.$$

- (iii) Tower diagnostics are obstruction-free on the overlap:

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0).$$

If Type IV diagnostics are invoked, the relevant tower comparison artifact on the overlap must be declared.

If the Overlap Gate passes on the declared overlap structure, local certificate records may be glued into a coherent global certificate record.

Definition 1.11 (B-Gate⁺ (repaired after-collapse gate)). Fix a filtered object F , a window W , and a threshold τ . We say that *B-Gate⁺ passes on $(F; W, \tau)$* if all of the following hold.

- (1) **Topological clause.** The monitored windowed collapsed profile vanishes in degree 1:

$$\text{Eval}_{1, W, \tau}(F) = 0.$$

Equivalently, if an admissible representative $C_{\tau, W}F$ is supplied,

$$\text{PH}_1(C_{\tau, W}F) = 0.$$

- (2) **Categorical clause.** If the Ext-clause is invoked, an admissible realization-compatible representative $C_{\tau, W}F$ is supplied, the eligible regime is declared and verified, and

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

- (3) **Type IV clause.** The monitored tower diagnostics are obstruction-free:

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0).$$

If these diagnostics are computed from a tower, the tower comparison artifact

$$\phi_{i, W, \tau}$$

is declared and auditable.

- (4) **Budget clause.** The safety margin dominates the ledger:

$$\text{gap}_\tau > \Sigma \delta.$$

Remark 1.12 (Interpretation of B-Gate⁺). B-Gate⁺ is an acceptance predicate for Core certification on a declared window and threshold. It is not a search score, not a probabilistic judgment, and not a domain-specific oracle. Any domain-specific interpretation of a passing or failing B-Gate⁺ belongs to an Interface theorem and must be declared separately.

Remark 1.13 (No representative, no Ext-clause). The persistence-level clause may be evaluated from $\text{Eval}_{1, W, \tau}(F)$. The Ext-clause may not be invoked unless a realization-compatible admissible representative $C_{\tau, W}F$ is supplied and the eligible regime is verified.

1.4. Failure landscape and the Type IV canon

Definition 1.14 (Failure landscape). We distinguish four failure types.

- **Type I.** Topological failure:

$$\text{Eval}_{1,W,\tau}(F) \text{ eq } 0$$

or equivalently $\text{PH}_1(C_{\tau,W}F) \text{ eq } 0$ when an admissible representative is supplied.

- **Type II.** Categorical failure in the eligible regime:

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) \text{ eq } 0.$$

- **Type III.** Operational or functorial instability under a declared finite-level pipeline operation, representative choice, windowing rule, or profile comparison.
- **Type IV.** Tower-level invisible obstruction: finite layers may appear admissible, while the declared collapsed tower comparison develops a monitored generic defect, detected by the infinite-bar kernel/cokernel diagnostics

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}).$$

Definition 1.15 (Tower comparison artifact). A tower comparison artifact for a monitored degree i , window W , and threshold τ is declared data specifying a comparison

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

Type IV diagnostics may be computed only after such a comparison artifact has been declared, unless an exact-admissible regime supplies it canonically.

Remark 1.16 (Type IV canonical convention). Throughout this document, the collapse diagnostics

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}})$$

are interpreted canonically as follows:

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0) \iff (\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0),$$

and

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \text{ eq } (0, 0) \iff (\mu_{\text{Collapse}}, u_{\text{Collapse}}) \text{ eq } (0, 0).$$

Thus “absence of Type IV” means absence of monitored generic / infinite-bar tower defect relative to the declared tower comparison artifact. It does not assert that every comparison map is an isomorphism on all finite-bar components unless additional hypotheses are declared.

Definition 1.17 (Windowed Collapse Zone). The *windowed Collapse Zone* on (W, τ) is

$$\mathfrak{C}_{W,\tau} := \{F \mid \text{Eval}_{1,W,\tau}(F) = 0 \wedge \text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \text{ whenever the Ext-clause is invoked}\}.$$

Under the Core bridge of Chapter 3, $\text{PH}_1 = 0$ implies $\text{Ext}^1 = 0$ in the eligible realization regime. No converse is globally asserted in the Core.

1.5. Minimal Manifest Axiom and proof-object discipline

Axiom 1.1 (Minimal Manifest Axiom). *A certified proof object is admissible only if it contains, at minimum, the following declared data.*

- (i) **Declared windows:** *the right-open windows used for certification and, when relevant, their overlap structure.*
- (ii) **Threshold policy:** *the chosen collapse threshold τ , or the declared τ -sweep and its interpretation policy.*
- (iii) **Collapse policy:** *whether the run uses*

barcode_level, tau_split, exact_admissible

collapse status.

- (iv) **Evaluation order:** *normally*

$$\mathbf{P}_i \longrightarrow W_W \mathbf{P}_i \longrightarrow T_\tau^{\text{bar}}(W_W \mathbf{P}_i).$$

- (v) **Representative policy:** *if $C_{\tau, W}F$ is used, the representative choice, monitored degrees, representative equivalence, and realization compatibility when relevant.*
- (vi) **Ledger policy:** *the quantale of budgets, the aggregation law, and the semantics of $\Sigma\delta$.*
- (vii) **Diagnostics:** *the monitored collapse diagnostics, including at least the data needed to determine $(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0)$ or $(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \text{ eq } (0, 0)$.*
- (viii) **Tower comparison artifacts:** *if Type IV diagnostics are invoked, the declared comparison artifact*

$$\phi_{i, W, \tau}$$

or the exact-admissible reason why it is canonical.

- (ix) **Gate verdicts:** *the pass/fail outcomes of $B\text{-Gate}^+$, and when gluing is used, the pass/fail outcomes of the Overlap Gate.*
- (x) **Artifact references:** *references to the concrete artifacts from which the above declarations and verdicts are reproducibly recoverable.*

Remark 1.18 (Why the manifest axiom belongs to the Core). The detailed storage format of proof objects may live in an external execution companion. However, the requirement that a proof object *have* declared windows, thresholds, collapse policy, evaluation order, representative policy, ledger semantics, diagnostics, tower comparison artifacts, gate verdicts, and artifact references is itself part of the Core audit discipline. Without this requirement, auditability would be delegated to an external implementation layer rather than built into the mathematical certification protocol.

Definition 1.19 (Certified proof object). A *certified proof object* is any object satisfying Axiom 1.1 together with the standing guard-rails of this chapter. A theorem in the present document may cite such an object as evidence only when the object is declared to be admissible in this sense.

1.6. Convergence manager on definable covers

Theorem 1.20 (Definable countable cover $\Rightarrow$ bounded global overlap error). *Let $\{W_n\}$ be a countable family of right-open definable windows with finite Čech depth K . If*

$$\sum_n \delta(W_n) < \infty$$

in the chosen quantale $\mathbb{V}$, then the cumulative overlap error is bounded, and the Overlap Gate may be checked globally on the covered domain by finite-depth gluing.

Remark 1.21 (Role of Theorem 1.20). Theorem 1.20 is a Core gluing theorem for certificate records. It is not a Search theorem. Its purpose is to certify when local Core verdicts may be assembled into a global certificate under quantale-summable budget control. All such gluing is performed on declared windowed collapsed profiles and their overlap records.

1.7. Explicit non-claims of Chapter 1

The following exclusions are part of the meaning of the Core fixed in this chapter.

- No unconditional global equivalence

$$\text{PH}_1 \Leftrightarrow \text{Ext}^1$$

is claimed.

- No claim is made that positive-threshold short-bar objects form a Serre subcategory in general.
- No claim is made that T_τ^{bar} is a global exact endofunctor on all of

$$\text{Pers}_k^{\text{cons}}.$$

- No claim is made that positive-threshold bar deletion is a global Serre localization or abelian reflector.
- No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is claimed.

- No global strict filtered lift

$$C_\tau : \text{FiltCh}(()k) \rightarrow \text{FiltCh}(()k)$$

is asserted by default.

- No identity

$$C_{\tau, W} F = (C_\tau F)|_W$$

is asserted without a declared window-compatibility condition.

- No tower comparison after positive-threshold deletion is automatic unless a tower comparison artifact or exact-admissible regime is declared.
- No Search artifact is treated as a Core proof object.
- No AI-generated proposal, ranking, map, heuristic, or confidence value is treated as a gate verdict.

- No problem-specific interpretation of a passing Core gate is asserted without an explicitly declared interface hypothesis.
- No scalar quantity, including Φ , replaces B-Gate⁺, tower diagnostics, or the budget condition.
- No reproducibility schema is fixed at the level of concrete serialization format inside the Core text. Only the minimal existence requirements of Axiom 1.1 are part of the Core.

1.8. Terminology and notation

Fix a base field k . We write Vect_k for finite-dimensional k -vector spaces and $\text{FiltCh}(()k)$ for finite-type filtered chain complexes. The degreewise persistence functor is denoted by $\mathbf{P}_i$. The barcode-level positive-threshold deletion operation is denoted by T_τ^{bar} . The window restriction operation is denoted by W_W . The repaired windowed collapsed evaluation is denoted by

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

The interleaving distance is denoted by d_{int} . All filtered (co)limits are interpreted objectwise in $[\mathbb{R}, \text{Vect}_k]$ and used only in the declared constructible regime or through declared windowed collapsed profile records.

1.9. Provenance and status labels

Existence of constructible interval decompositions and barcode stability belongs to standard persistence theory. The repaired positive-threshold collapse discipline

$$T_\tau^{\text{bar}}, \quad \text{Eval}_{i,W,\tau}, \quad C_{\tau,W}F$$

is part of the present AK framework. It replaces the stronger earlier language of global positive-threshold Serre localization.

Derived, sheaf-theoretic, higher-categorical, arithmetic, PDE, and Search-layer realizations may appear later as interface or extension material, but they do not enlarge the Core fixed in this chapter.

Status labels for this chapter. *Core mathematical theorem:* Theorem 1.3, Theorem 1.20.

Operational policy / audit axiom: Policy 1, Definitions 1.4, 1.5, 1.6, 1.7, 1.10, 1.11, 1.15, Axiom 1.1.

Core doctrine: Remark 1.2, Remark 1.9, Remark 1.12, Remark 1.13, Remark 1.16.

Explicit non-claims: Section 1.7.

2 Chapter 2: Constructible Persistence, Barcode Collapse, and Windowed Evaluation

2.0. Scope and correction of positive-threshold collapse

This chapter fixes the persistence-theoretic foundation used by the AK Core after the v18.1 repair. It replaces the stronger positive-threshold Serre-localization language by a more conservative and auditable barcode-level collapse discipline.

The central rule of this chapter is:

For a positive threshold $\tau > 0$, short-bar deletion is used in the Core as a barcode-level, idempotent, threshold-monotone, and stable collapse operation. It is not asserted, in the general Core regime, to be an exact Serre reflector on all of $\text{Pers}_k^{\text{cons}}$.

This correction is necessary because, for $\tau > 0$, the class of persistence modules built only from bars of length at most τ is not extension-closed in general. Therefore it cannot be treated as a Serre subcategory without additional hypotheses.

The purpose of this chapter is to define precisely:

- (i) the constructible one-parameter persistence regime;
- (ii) barcode normal forms and bar lengths;
- (iii) the barcode-level threshold deletion operation T_τ^{bar} ;
- (iv) the fixed windowed evaluation order used by the Core;
- (v) the exact-admissible regimes in which stronger functorial or exact statements may be recovered;
- (vi) the consequences for $B\text{-Gate}^+$, filtered representatives, and Type IV diagnostics.

Remark 2.1 (Repair relative to the previous formulation). Earlier formulations described positive-threshold short-bar deletion as an exact Serre reflector. In the general constructible persistence category this is too strong. The repaired Core keeps the operational and metric content needed by AK—threshold deletion, stability, windowed evaluation, and auditability—while removing the unsupported global exactness claim.

2.1. The constructible persistence setting

Fix a field k . Let

$$\text{Pers}_k := [(\mathbb{R}, \leq), \text{Vect}_k]$$

be the category of one-parameter persistence modules over k , indexed by the ordered real line.

Definition 2.2 (Constructible persistence module). A persistence module $M \in \text{Pers}_k$ is called *constructible* if:

- (i) $M(t)$ is finite-dimensional over k for every $t \in \mathbb{R}$;
- (ii) on every bounded interval $I \subset \mathbb{R}$, there exists a finite critical set outside which all structure maps are isomorphisms;
- (iii) there is no wild accumulation of critical parameters inside bounded intervals.

We denote the full subcategory of constructible persistence modules by

$$\text{Pers}_k^{\text{cons}} \subset \text{Pers}_k.$$

Remark 2.3 (Working regime). All Core statements in this chapter are made inside $\text{Pers}_k^{\text{cons}}$. When object-wise limits or colimits are used in $[\mathbb{R}, \text{Vect}_k]$, the result must either be verified to lie again in the constructible regime or be processed through a declared windowed evaluation protocol. No statement in this chapter applies to wild, nonconstructible, infinite-rank persistence data.

Definition 2.4 (Finite-type filtered chain complex). Let $\text{Ch}^b(k)$ be the category of bounded chain complexes of finite-dimensional k -vector spaces. A *finite-type filtered chain complex* is a pair

$$F = (C_\bullet, \{F^t C_\bullet\}_{t \in \mathbb{R}})$$

such that:

- (i) $F^t C_\bullet \subseteq F^{t'} C_\bullet$ for $t \leq t'$;
- (ii) the filtration is exhaustive and left-bounded;
- (iii) for each homological degree i , the persistence module

$$\mathbf{P}_i(F) : \mathbb{R} \rightarrow \text{Vect}_k, \quad t \mapsto H_i(F^t C_\bullet)$$

belongs to $\text{Pers}_k^{\text{cons}}$.

We denote the category of such objects, with filtration-preserving chain maps, by

$$\text{FiltCh}(\cdot)k.$$

Remark 2.5 (Persistence layer first). The Core reads filtered objects only through their monitored persistence profiles

$$\mathbf{P}_i(F).$$

Filtered-level statements are interpreted through the persistence layer unless an additional filtered representative policy is declared.

2.2. Barcode normal forms

Definition 2.6 (Interval module). For an interval $I \subseteq \mathbb{R}$, the interval module k_I is the persistence module defined by

$$k_I(t) = \begin{cases} k, & t \in I, \\ 0, & t \notin I, \end{cases}$$

with structure maps equal to the identity whenever both source and target lie in I , and zero otherwise.

Definition 2.7 (Bar and barcode). An interval summand k_I is called a *bar*. A barcode of a constructible persistence module M is an interval decomposition

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda},$$

with locally finite barcode support on bounded intervals. We denote the corresponding multiset of bars by

$$\mathcal{B}(M) = \{I_\lambda\}_{\lambda \in \Lambda}.$$

Theorem 2.8 (Constructible interval decomposition). *Every object $M \in \text{Pers}_k^{\text{cons}}$ admits a locally finite interval decomposition*

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda}.$$

The barcode is unique up to permutation of interval summands.

Proof sketch. This is the standard interval decomposition theorem for pointwise finite-dimensional constructible one-parameter persistence modules over a field. On bounded intervals, constructibility reduces the module to a finite representation of a linearly ordered finite poset. Such representations decompose into interval representations. Local finiteness yields the global constructible barcode, and uniqueness follows from the Krull–Schmidt property in the interval-decomposable regime. $\square$

Definition 2.9 (Length of a bar). Let $I \subset \mathbb{R}$ be a bounded interval with endpoints $a < b$. Its length is

$$\ell(I) := b - a.$$

If I is unbounded above, it is called an infinite bar and is assigned length $+\infty$ for thresholding purposes.

Remark 2.10 (Endpoint independence). The length $\ell(I)$ depends only on the endpoint values, not on whether endpoints are open, closed, or half-open. Hence any threshold rule based solely on $\ell(I)$ is independent of harmless endpoint bookkeeping conventions.

2.3. Positive-threshold short bars are not Serre in general

The following observation is the key repair point of v18.1.

Proposition 2.11 (Positive-threshold short-bar objects are not Serre in general). *Let $\tau > 0$. The class of constructible persistence modules whose barcode summands all have finite length at most τ is not extension-closed in general. Consequently, it is not a Serre subcategory of $\text{Pers}_k^{\text{cons}}$ in general.*

Proof. It suffices to give a counterexample. Take $\tau = 1$ and consider the short exact sequence of interval modules

$$0 \longrightarrow k_{[1,2)} \longrightarrow k_{[0,2)} \longrightarrow k_{[0,1)} \longrightarrow 0.$$

The subobject $k_{[1,2)}$ has length 1, and the quotient $k_{[0,1)}$ also has length 1. However, the middle object $k_{[0,2)}$ has length 2. Thus an extension of two short-bar objects can contain a longer bar. Therefore the class of bars of length at most τ is not extension-closed in general. $\square$

Corollary 2.12 (No global exact reflector claim). *For $\tau > 0$, the operation of deleting bars of length at most τ is not asserted in the general Core to arise from an exact Serre localization or an abelian exact reflector on all of $\text{Pers}_k^{\text{cons}}$.*

Remark 2.13 (Why the correction matters). If positive-threshold deletion were treated as an exact Serre reflector without additional hypotheses, then exactness, functoriality, tower comparison maps, and filtered lifts would all inherit unsupported categorical strength. The repaired formulation keeps the barcode-level operation but requires stronger claims to be explicitly justified by exact-admissible hypotheses.

2.4. Barcode-level threshold deletion

Definition 2.14 (Barcode threshold deletion). Let $M \in \text{Pers}_k^{\text{cons}}$ have barcode decomposition

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda}.$$

For $\tau > 0$, define the *barcode-level threshold deletion* of M by

$$T_\tau^{\text{bar}}(M) := \bigoplus_{\lambda: \ell(I_\lambda) > \tau \text{ or } \ell(I_\lambda) = +\infty} k_{I_\lambda}.$$

Equivalently, T_τ^{bar} deletes precisely the finite bars of length at most τ .

Remark 2.15 (Status of T_τ^{bar}). In the general Core regime, T_τ^{bar} is an operation on barcode normal forms, and hence on isomorphism classes of constructible persistence modules. It is not, without further hypotheses, asserted to be a strict exact endofunctor on all morphisms of $\text{Pers}_k^{\text{cons}}$.

Proposition 2.16 (Basic properties of barcode threshold deletion). *For every $\tau > 0$, the operation T_τ^{bar} satisfies:*

(i) *endpoint independence;*

(ii) *idempotence:*

$$T_\tau^{\text{bar}} T_\tau^{\text{bar}}(M) \cong T_\tau^{\text{bar}}(M);$$

(iii) *infinite-bar preservation;*

(iv) *deletion monotonicity in the threshold.*

Proof. All statements follow directly from the barcode definition. Deleting all finite bars of length at most τ twice has the same effect as deleting them once. Infinite bars have length $+\infty$, hence are never deleted. Endpoint choices do not affect bar length. Finally, increasing the threshold can only delete additional finite bars. $\square$

Proposition 2.17 (Threshold composition). *For $\tau, \sigma > 0$,*

$$T_\sigma^{\text{bar}} T_\tau^{\text{bar}}(M) \cong T_{\max\{\tau, \sigma\}}^{\text{bar}}(M)$$

at the barcode level.

Proof. Applying T_τ^{bar} deletes bars of length at most τ . Applying T_σ^{bar} afterwards deletes bars of length at most σ among those remaining. The total effect is deletion of all finite bars of length at most $\max\{\tau, \sigma\}$. $\square$

Definition 2.18 (Threshold comparison record). If $0 < \tau \leq \sigma$, then σ represents a stronger collapse threshold. At the barcode-profile level, the retained bars of $T_\sigma^{\text{bar}}(M)$ form a submultiset of the retained bars of $T_\tau^{\text{bar}}(M)$. The Core records this relation as the threshold comparison

$$T_\tau^{\text{bar}}(M) \rightsquigarrow T_\sigma^{\text{bar}}(M).$$

This is a comparison of retained barcode profiles, not a global exact natural transformation unless an exact-admissible regime has been declared.

Remark 2.19 (Direction of threshold comparison). The direction is from weaker collapse to stronger collapse:

$$0 < \tau \leq \sigma \implies T_\tau^{\text{bar}} \rightsquigarrow T_\sigma^{\text{bar}}.$$

This convention is used throughout threshold sweeps and audit records.

2.5. Stability at the barcode metric level

Let d_{int} denote the interleaving distance. In the constructible one-parameter regime over a field, it agrees with the bottleneck distance on barcodes under the usual isometry theorem.

Theorem 2.20 (Barcode non-expansiveness). *For all $M, N \in \text{Pers}_k^{\text{cons}}$,*

$$d_{\text{int}}(T_\tau^{\text{bar}}(M), T_\tau^{\text{bar}}(N)) \leq d_{\text{int}}(M, N)$$

at the barcode metric level.

Proof sketch. Under the barcode isometry theorem, the interleaving distance is computed as the bottleneck distance between barcodes. Removing all bars of length at most τ from both barcodes cannot force a larger matching cost between the retained barcodes. Hence the threshold-deletion operation is non-expansive. $\square$

Remark 2.21 (Metric content retained by the repair). The v18.1 repair removes unsupported global exactness, but it retains the metric property needed by the after-collapse audit doctrine: threshold deletion does not amplify barcode-level interleaving error.

2.6. Window restriction and non-commutation

Definition 2.22 (Right-open window). A domain window is a right-open interval

$$W = [u, u') \subset \mathbb{R}.$$

A family of such windows is MECE if it is mutually exclusive and collectively exhaustive on the declared domain range.

Definition 2.23 (Window restriction). For $W = [u, u')$, define the window restriction

$$W_W(M)$$

by restricting M to W , and, when a global persistence module is required, extending by zero outside the declared window.

At the barcode level, W_W clips each bar I to

$$I \cap W,$$

discarding empty intersections.

Proposition 2.24 (Window restriction is barcode-stable). *At the barcode metric level, W_W is non-expansive:*

$$d_{\text{int}}(W_W(M), W_W(N)) \leq d_{\text{int}}(M, N).$$

Proof sketch. Clipping bars to a fixed right-open interval cannot increase the bottleneck cost of an optimal matching. Equivalently, restriction to a fixed window does not create larger displacement between corresponding bar endpoints. $\square$

Proposition 2.25 (Window restriction and threshold deletion do not commute in general). *In general,*

$$W_W T_\tau^{\text{bar}}(M) \not\cong T_\tau^{\text{bar}} W_W(M).$$

Proof. Let $M = k_{[0,10]}$, let $W = [0, 1)$, and take $\tau = 2$. Since the global bar has length $10 > \tau$, it survives T_τ^{bar} . Therefore $W_W T_\tau^{\text{bar}}(M) \cong k_{[0,1]}$.

On the other hand, $W_W(M) \cong k_{[0,1]}$, whose length is $1 \leq \tau$. Thus

$$T_\tau^{\text{bar}} W_W(M) = 0.$$

Hence the two operations do not commute in general. $\square$

Remark 2.26 (Core consequence). No unconditional crop-collapse commutation is part of the Core. Whenever a theorem or gate requires windowed collapse data, the evaluation order must be declared.

2.7. Core windowed evaluation order

Definition 2.27 (Windowed collapsed persistence evaluation). Let $F \in \text{FiltCh}(()k)$, let i be a monitored degree, let W be a declared right-open window, and let $\tau > 0$. The Core windowed collapsed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Remark 2.28 (Fixed order). The expression

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F))$$

means:

$$F \mapsto \mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Thus the Core first restricts to the declared window and only then applies threshold deletion inside that window.

Definition 2.29 (Window-adapted commutation condition). A pair (M, W) is called τ -window-adapted if every finite bar of M that survives T_{τ}^{bar} either:

- (i) remains of clipped length $> \tau$ after restriction to W ; or
- (ii) is disjoint from W .

Proposition 2.30 (Conditional commutation). *If (M, W) is τ -window-adapted, then*

$$W_W T_{\tau}^{\text{bar}}(M) \cong T_{\tau}^{\text{bar}} W_W(M)$$

at the barcode level.

Proof. Under the τ -window-adapted condition, no globally surviving bar is shortened by window restriction into the deleted length range $\leq \tau$. Therefore the set of bars retained after applying window restriction and threshold deletion is the same in either order. $\square$

Remark 2.31 (Default policy). The default Core policy is not to assume τ -window-adaptedness. Therefore all gate-relevant quantities are evaluated using $\text{Eval}_{i,W,\tau}$. Conditional commutation may be used only when explicitly declared in the run manifest or in a theorem hypothesis.

2.8. Exact-admissible regimes

The general Core does not treat T_{τ}^{bar} as a global exact functor. Nevertheless, stronger exact calculus may be available under separately declared hypotheses.

Definition 2.32 (τ -split object). An object $M \in \text{Pers}_k^{\text{cons}}$ is called τ -split if it is equipped with a declared decomposition

$$M \cong M_{\leq \tau} \oplus M_{> \tau}$$

such that:

- (i) every finite bar of $M_{\leq \tau}$ has length at most τ ;
- (ii) every finite bar of $M_{> \tau}$ has length greater than τ ;
- (iii) all infinite bars, if any, lie in $M_{> \tau}$;

(iv) the decomposition is part of the declared data, not merely inferred informally.

Definition 2.33 (τ -split morphism). A morphism

$$f : M \rightarrow N$$

between τ -split objects is called τ -split if it respects the declared decompositions:

$$f(M_{\leq \tau}) \subseteq N_{\leq \tau}, \quad f(M_{> \tau}) \subseteq N_{> \tau}.$$

Definition 2.34 (τ -split regime). A τ -split regime is a full declared working subcategory or workflow domain in which all objects and all morphisms under consideration are τ -split in the above sense.

Proposition 2.35 (Functorial deletion in a τ -split regime). *Inside a declared τ -split regime, the assignment*

$$M \mapsto M_{> \tau}$$

is functorial and agrees with $T_{\tau}^{\text{bar}}(M)$ on barcode profiles.

Proof. Since every morphism respects the declared short/long decomposition, it restricts to a morphism

$$M_{> \tau} \rightarrow N_{> \tau}.$$

Therefore the assignment $M \mapsto M_{> \tau}$ is functorial in the τ -split regime. By construction, $M_{> \tau}$ consists exactly of the bars retained by T_{τ}^{bar} . $\square$

Definition 2.36 (Exact-admissible collapse regime). An exact-admissible collapse regime is a declared τ -split regime in which the retained-long-bar assignment

$$M \mapsto M_{> \tau}$$

is additionally verified to preserve the exact sequences used in the relevant Core argument.

Remark 2.37 (Exactness is conditional). Exactness of positive-threshold collapse is not a global Core theorem. It is a conditional property available only in an exact-admissible regime. Whenever such a regime is used, the proof object must declare:

- (i) the τ -split decomposition policy;
- (ii) the morphism compatibility policy;
- (iii) the exact sequences being tested;
- (iv) the artifact or proof establishing exact preservation.

2.9. Filtered representatives of windowed collapse

Because T_{τ}^{bar} is treated in the general Core as a barcode-level operation, filtered-level collapse must be phrased in terms of representatives.

Definition 2.38 (Windowed collapsed persistence profile). For $F \in \text{FiltCh}(()k)$, monitored degree i , window W , and threshold τ , define the monitored windowed collapsed profile by

$$\mathcal{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Definition 2.39 (Admissible filtered representative). Let $F \in \text{FiltCh}((k))$, let W be a declared window, and let $\tau > 0$. A filtered object

$$C_{\tau,W}F$$

is called an admissible filtered representative of the windowed collapse of F if, for every monitored degree i ,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F)$$

at the persistence barcode level, up to the declared equivalence policy.

Remark 2.40 (No global strict lift by default). The general Core does not assert a strict global endofunctor

$$C_\tau : \text{FiltCh}((k)) \rightarrow \text{FiltCh}((k))$$

realizing positive-threshold collapse in every degree and on every morphism. The default object is the windowed representative $C_{\tau,W}F$, whose existence and properties must be declared in the proof object or implementation record.

Definition 2.41 (Representative equivalence). Two admissible representatives $C_{\tau,W}F$ and $C'_{\tau,W}F$ are considered equivalent for Core gate purposes if their monitored persistence profiles agree:

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathbf{P}_i(C'_{\tau,W}F)$$

for all monitored degrees i , under the declared barcode equivalence policy.

Remark 2.42 (Filtered quasi-isomorphism). When filtered quasi-isomorphism data are available, representative equivalence may be strengthened to filtered quasi-isomorphism. However, filtered quasi-isomorphism is not required merely to define the windowed collapsed barcode profile. It becomes necessary only when the argument uses filtered-level or derived-level constructions.

2.10. Consequences for $B\text{-Gate}^+$

The repaired Chapter 2 affects how the Core gate is read.

Definition 2.43 (Windowed B -side input). The B -side input for a monitored degree i , window W , and threshold τ is the collapsed profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)),$$

or an admissible filtered representative $C_{\tau,W}F$ realizing this profile.

Definition 2.44 (Windowed topological clause). The topological clause of $B\text{-Gate}^+$ on $(F; W, \tau)$ is read as

$$PH_1(C_{\tau,W}F) = 0,$$

or equivalently,

$$\text{Eval}_{1,W,\tau}(F) = 0$$

at the monitored persistence-profile level, when the representative realizes the declared profile.

Definition 2.45 (Windowed categorical clause). The categorical clause of $B\text{-Gate}^+$ may be invoked only when an admissible filtered representative $C_{\tau,W}F$ is supplied and the eligible realization condition holds:

$$R(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-mod}).$$

Only then is the Ext-clause

$$\text{Ext}^1(R(C_{\tau,W}F), k) = 0$$

well-typed for the Core bridge.

Remark 2.46 (PH before Ext). The barcode-level profile $\text{Eval}_{1,W,\tau}(F)$ is the primary persistence datum. The Ext-clause is not a replacement for it. Ext is read only through an eligible representative and the one-way bridge established later.

2.11. Tower comparison and Type IV diagnostics

The repaired collapse discipline also changes how tower diagnostics are introduced. Since T_τ^{bar} is not a global strict exact functor in the general Core, tower comparison maps must be declared rather than assumed.

Definition 2.47 (Windowed collapsed tower profile). Let

$$F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow \dots$$

be a directed tower in the declared filtered or persistence regime. For a monitored degree i , window W , and threshold τ , the associated windowed collapsed tower profile is the sequence of barcode profiles

$$\mathcal{P}_{i,W,\tau}(F_0), \mathcal{P}_{i,W,\tau}(F_1), \mathcal{P}_{i,W,\tau}(F_2), \dots$$

provided the comparison data between these profiles are declared.

Definition 2.48 (Tower comparison artifact). A tower comparison artifact for (i, W, τ) is declared data specifying a comparison

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty)$$

between the collapsed profile of the tower and the collapsed profile of its declared apex or limiting object.

Remark 2.49 (No automatic tower comparison). In the general Core, the comparison map $\phi_{i,W,\tau}$ is not automatic. It is part of the proof object or run manifest. In exact-admissible regimes, such maps may be canonical; otherwise they must be explicitly constructed, logged, and audited.

Definition 2.50 (Type IV diagnostic input). Type IV diagnostics are computed only after a tower comparison artifact has been declared. The monitored kernel and cokernel defects are then read from

$$\ker(\phi_{i,W,\tau}), \quad \text{coker}(\phi_{i,W,\tau}),$$

in the declared persistence-profile or exact-admissible regime.

Remark 2.51 (Type IV remains Core, but its input is declared). The Type IV diagnostic remains part of the Core gate discipline. The repair only changes its input discipline: the tower comparison is no longer silently inherited from a global exact collapse functor. It must be either canonical in an exact-admissible regime or explicitly declared as an artifact.

2.12. Minimal manifest consequences

A certified proof object using Chapter 2 must record the following additional data.

Definition 2.52 (Collapse policy manifest entry). A collapse policy manifest entry contains:

- (i) the threshold τ ;
- (ii) the monitored degrees i ;
- (iii) the window W ;
- (iv) the evaluation order, normally

$$\mathbf{P}_i \rightarrow W_W \rightarrow T_\tau^{\text{bar}};$$

(v) the collapse status:

barcode_level, tau_split, exact_admissible;

(vi) whether an admissible filtered representative $C_{\tau,W}F$ is supplied;

(vii) whether tower comparison artifacts are required and supplied.

Remark 2.53 (Audit consequence). A proof object that simply states “apply T_τ ” is incomplete under v18.1. It must state whether T_τ means barcode-level threshold deletion, a τ -split functorial model, or an exact-admissible collapse regime.

2.13. Operating summary

For each monitored object F , degree i , window W , and threshold τ , the repaired Chapter 2 requires the following sequence:

$$F \mapsto \mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)) = \text{Eval}_{i,W,\tau}(F).$$

If a filtered or derived argument is needed, one must additionally supply an admissible representative

$$C_{\tau,W}F$$

satisfying

$$\mathbf{P}_i(C_{\tau,W}F) \cong \text{Eval}_{i,W,\tau}(F)$$

for monitored degrees.

If a tower diagnostic is needed, one must supply a tower comparison artifact

$$\phi_{i,W,\tau}.$$

If exactness or strict functoriality is needed, one must work in a declared τ -split or exact-admissible regime.

2.14. Explicit non-claims

The following are not asserted in this chapter.

(i) No claim is made that positive-threshold short-bar objects form a Serre subcategory in general.

(ii) No claim is made that T_τ^{bar} is a global exact endofunctor on all of $\text{Pers}_k^{\text{cons}}$.

(iii) No claim is made that T_τ^{bar} is a global abelian reflector or Serre localization.

(iv) No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is asserted.

(v) No global strict filtered lift

$$C_\tau : \text{FiltCh}(()k) \rightarrow \text{FiltCh}(()k)$$

is asserted by default.

(vi) No tower comparison map after threshold deletion is automatic unless a tower comparison artifact or exact-admissible regime is declared.

- (vii) No Ext-clause is well-typed without an admissible filtered representative and an eligible realization condition.
- (viii) No Search artifact, advisory invariant, spectral indicator, or AI-generated ranking replaces the windowed collapsed persistence evaluation.

2.15. Provenance and status

The constructible persistence setting, interval decomposition, barcode uniqueness, bar-length convention, and bottleneck/interleaving stability are standard persistence-theoretic facts in the one-parameter constructible regime over a field.

The positive-threshold repair, the barcode-level interpretation of T_τ^{bar} , the fixed windowed evaluation order, and the requirement of tower comparison artifacts are part of the present AK v18.1 Core discipline.

Status labels for this chapter. *Core mathematical theorem:* Theorem 2.8, Theorem 2.20.

Core correction theorem/proposition: Proposition 2.11, Corollary 2.12, Proposition 2.25.

Core definitions: Definition 2.2, Definition 2.4, Definition 2.7, Definition 2.14, Definition 2.27, Definition 2.39, Definition 2.48.

Operational policy: Definition 2.52, Remark 2.28, Remark 2.31, Remark 2.49.

Exact-admissible optional regime: Definition 2.32, Definition 2.33, Definition 2.34, Definition 2.36, Proposition 2.35.

Explicit exclusions: Section 2.14.

3 Chapter 3: Eligibility, Ext-edge, and the One-Way Bridge $\text{PH}_1 \Rightarrow \text{Ext}^1$

3.1. Scope and standing discipline

All statements in this chapter are made in the repaired Core regime fixed by Chapters 1–2 and Appendices A–B.

In particular:

- (i) persistence data live in constructible one-parameter persistence over a field k ;
- (ii) positive-threshold collapse is read as the barcode-level operation T_τ^{bar} ;
- (iii) gate-relevant persistence data are evaluated on right-open windows $W = [u, u')$;
- (iv) the Core evaluation order is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F));$$

- (v) filtered objects $C_{\tau,W}F$ are admissible representatives of the windowed collapsed profile, not global collapse functors;
- (vi) any Ext-level statement requires an admissible realization-compatible representative and the eligible realization regime.

The purpose of the present chapter is narrowly delimited:

- (i) to state the eligible realization regime;
- (ii) to isolate the Ext^1 -edge in amplitude $[-1, 0]$;

(iii) to prove the one-way implication

$$\text{PH}_1 \Rightarrow \text{Ext}^1.$$

Remark 3.1 (What this chapter does not do). This chapter does *not* assert any converse implication

$$\text{Ext}^1 \Rightarrow \text{PH}_1.$$

It also does not use CCC authorization, spectral-gap hypotheses, Bridge Programs, validity maps, advisory invariants, or search-layer reports. Any such reverse-use doctrine belongs outside the Core, in later interface, extension, or search-side material.

3.2. Windowed collapsed persistence profiles

Fix $F \in \text{FiltCh}(\cdot)k$, a monitored degree i , a right-open window $W = [u, u')$, and a threshold $\tau > 0$.

Definition 3.2 (Windowed collapsed persistence profile). The *windowed collapsed persistence profile* of F in degree i is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)) \in \text{Pers}_k^{\text{cons}}$$

at the barcode-profile level.

Remark 3.3 (Fixed order). The order in Definition 3.2 is part of the Core. The window restriction is applied before threshold deletion:

$$F \mapsto \mathbf{P}_i(F) \mapsto \mathbb{W}_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)).$$

No unconditional identity

$$\mathbb{W}_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} \mathbb{W}_W$$

is used in this chapter.

Definition 3.4 (Admissible filtered representative). An *admissible filtered representative* of the windowed collapsed profile is a filtered object

$$C_{\tau,W}F \in \text{FiltCh}(\cdot)k$$

such that, for every monitored degree i ,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F))$$

under the declared representative equivalence policy.

Remark 3.5 (Why $C_{\tau,W}$ is used). Writing $C_\tau F|_W$ without qualification can conceal an order issue: window restriction may change bar lengths and therefore need not commute with length-threshold deletion. The notation $C_{\tau,W}F$ records the repaired Core convention that the gate-relevant object is the windowed collapsed profile

$$\text{Eval}_{i,W,\tau}(F).$$

Remark 3.6 (Representative, not collapse). The representative $C_{\tau,W}F$ does not define the collapse operation. The collapse data are already defined at the barcode-profile level by $\text{Eval}_{i,W,\tau}(F)$. The representative is used only when filtered-chain or derived-category constructions are needed.

3.3. Eligible realization regime

The one-way bridge is stated only for representatives satisfying the following hypotheses.

(B1) Constructible monitored input. $F \in \text{FiltCh}(\cdot)k$ is finite type, and $\mathbf{P}_i(F) \in \text{Pers}_k^{\text{cons}}$ for the monitored degrees. The windowed collapsed profiles

$$\text{Eval}_{i,w,\tau}(F)$$

are defined in the repaired Core sense.

(B2) Admissible representative. An admissible filtered representative

$$C_{\tau,w}F$$

is supplied, satisfying

$$\mathbf{P}_i(C_{\tau,w}F) \cong \text{Eval}_{i,w,\tau}(F)$$

for all monitored degrees i .

(B3) Realization into the two-term derived regime. There exists a realization functor

$$\mathcal{R} : \text{FiltCh}(\cdot)k \rightarrow D^b(k\text{-mod})$$

such that

$$\mathcal{R}(C_{\tau,w}F) \in D^{[-1,0]}(k\text{-mod}).$$

(B4) Edge-realization condition. The -1 -st cohomological edge of

$$\mathcal{R}(C_{\tau,w}F)$$

is realized by the residual degree-1 windowed collapsed persistence edge. Concretely, there is a declared edge isomorphism or edge identification

$$H^{-1}(\mathcal{R}(C_{\tau,w}F)) \cong \mathcal{H}_1^{\tau,w}(F),$$

where $\mathcal{H}_1^{\tau,w}(F)$ is the vector-space edge represented by

$$\text{Eval}_{1,w,\tau}(F).$$

In particular,

$$\text{Eval}_{1,w,\tau}(F) = 0 \implies H^{-1}(\mathcal{R}(C_{\tau,w}F)) = 0.$$

(B5) Field coefficient regime. All realized objects are taken in $D^b(k\text{-mod})$, where k -modules are semisimple.

Definition 3.7 (Eligible regime). A triple $(F; W, \tau)$, together with an admissible representative $C_{\tau,w}F$, is said to be *eligible* if it satisfies (B1)–(B5). Whenever B-Gate⁺ invokes the Ext¹-clause on a window, eligibility must be declared and checked.

Remark 3.8 (Role of eligibility). Eligibility is not a cosmetic side condition. It is the exact point at which the Ext¹-edge becomes available and becomes linked to the windowed collapsed degree-1 persistence edge. Without eligibility, the Ext¹-part of the Core gate is not licensed.

Remark 3.9 (No representative, no Ext-clause). The persistence-level clause of B-Gate⁺ may be evaluated from

$$\text{Eval}_{1,w,\tau}(F).$$

The Ext-clause may not be invoked unless an admissible realization-compatible representative

$$C_{\tau,w}F$$

is supplied and the eligible regime is verified.

3.4. The realized edge object

Definition 3.10 (Eligible realized edge object). For an eligible triple $(F; W, \tau)$, define the realized edge object

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-mod}).$$

All Ext^1 -statements in this chapter are statements about $A_{F,W,\tau}$.

Remark 3.11 (Filtered representative and realization order). The Core first fixes the windowed collapsed persistence profile

$$\text{Eval}_{i,W,\tau}(F).$$

Then, only when needed, it chooses an admissible filtered representative $C_{\tau,W}F$. The realization $\mathcal{R}$ is applied to that representative. Thus the order is:

$$F \longmapsto \text{Eval}_{i,W,\tau}(F) \longmapsto C_{\tau,W}F \longmapsto \mathcal{R}(C_{\tau,W}F).$$

3.5. The Ext^1 -edge

Lemma 3.12 (Splitting over a field). *Let $A \in D^b(k\text{-mod})$. Since $k\text{-mod}$ is semisimple, A is isomorphic in $D^b(k\text{-mod})$ to the direct sum of its cohomology objects:*

$$A \cong \bigoplus_{j \in \mathbb{Z}} H^j(A)[-j].$$

Proof. Over a field, every short exact sequence of k -vector spaces splits. Hence every bounded complex is quasi-isomorphic to its cohomology with zero differential. This gives the stated decomposition in the derived category. $\square$

Theorem 3.13 (Ext^1 -edge identification). *Let $A \in D^{[-1,0]}(k\text{-mod})$. Then there is a natural identification*

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

Proof. By Lemma 3.12,

$$A \cong H^{-1}(A)[1] \oplus H^0(A).$$

Using

$$\text{Ext}^1(A, k) = \text{Hom}_{D^b(k\text{-mod})}(A, k[1]),$$

we obtain

$$\text{Hom}(A, k[1]) \cong \text{Hom}(H^{-1}(A)[1], k[1]) \oplus \text{Hom}(H^0(A), k[1]).$$

The first summand is

$$\text{Hom}_k(H^{-1}(A), k).$$

The second summand vanishes because $k\text{-mod}$ is semisimple and hence

$$\text{Ext}_k^1(H^0(A), k) = 0.$$

Therefore

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

$\square$

Corollary 3.14 (Vanishing criterion at the edge). *If $A \in D^{[-1,0]}(k\text{-mod})$ satisfies*

$$H^{-1}(A) = 0,$$

then

$$\text{Ext}^1(A, k) = 0.$$

Proof. Immediate from Theorem 3.13. □

Remark 3.15 (Why only the -1 -edge matters). In the eligible two-term regime, the Ext^1 -obstruction is controlled by the -1 -st cohomological edge. This is the only categorical input needed for the one-way bridge. No reverse information is extracted from this fact in the present chapter.

3.6. The one-way bridge $\text{PH}_1 \Rightarrow \text{Ext}^1$

Theorem 3.16 (One-way bridge). *Let $(F; W, \tau)$ be eligible. If*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

equivalently, for an admissible representative,

$$\text{PH}_1(C_{\tau,W}F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Let

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F).$$

By eligibility,

$$A_{F,W,\tau} \in D^{[-1,0]}(k\text{-mod})$$

and the -1 -st cohomology edge is realized by the residual degree-1 windowed collapsed persistence edge.

The hypothesis

$$\text{Eval}_{1,W,\tau}(F) = 0$$

therefore implies

$$H^{-1}(A_{F,W,\tau}) = 0.$$

By Corollary 3.14,

$$\text{Ext}^1(A_{F,W,\tau}, k) = 0.$$

This is exactly

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

□

Remark 3.17 (Core meaning of the bridge). Theorem 3.16 is a discharge theorem. It says that, in the eligible regime, topological vanishing of the repaired windowed collapsed degree-1 persistence profile is sufficient to discharge the categorical Ext^1 -clause of the Core gate. It does not say that categorical vanishing reconstructs topological vanishing.

3.7. Minimal E_1 -first rule compatible with the repaired Core

The present chapter retains only the minimal form of the E_1 -first doctrine needed for Core bookkeeping.

Definition 3.18 (Windowed first-length functional). Let

$$E_1(F; W, \tau)$$

denote the sum of the lengths of all residual degree-1 bars in the windowed collapsed profile

$$\text{Eval}_{1,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_1(F)).$$

Definition 3.19 (Minimal E_1 -first rule). The Core uses E_1 only as an operational detector of immediate post-collapse degree-1 vanishing:

$$E_1(F; W, \tau) = 0 \implies \text{Eval}_{1,W,\tau}(F) = 0.$$

If an admissible representative $C_{\tau,W}F$ is supplied, this may also be read as

$$E_1(F; W, \tau) = 0 \implies \text{PH}_1(C_{\tau,W}F) = 0.$$

No converse, no local equivalence theorem, and no reverse bridge are asserted here.

Remark 3.20 (Why E_1 is retained). The point of retaining E_1 is operational minimality. It provides a measurable after-collapse vanishing witness compatible with B-Gate⁺. It is not introduced as an independent theorem schema and does not replace PH_1 , $\text{Eval}_{1,W,\tau}$, or Ext^1 .

3.8. Gate integration

The bridge of Theorem 3.16 is used only in the repaired after-collapse Core gate of Chapter 1.

Proposition 3.21 (B-Gate⁺ discharge of the Ext^1 -clause). *Let $(F; W, \tau)$ be eligible. Assume*

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

Then the Ext^1 -clause of B-Gate⁺ on $(F; W, \tau)$ is satisfied:

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. This is exactly Theorem 3.16. □

Remark 3.22 (What still remains to pass B-Gate⁺). The one-way bridge discharges only the Ext^1 -clause. A full pass of B-Gate⁺ still requires the separate verification of:

$$\text{Eval}_{1,W,\tau}(F) = 0, \quad (\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0), \quad \text{gap}_\tau > \Sigma\delta.$$

If Type IV diagnostics are invoked, the relevant tower comparison artifact must also be declared and auditable.

3.9. Representative invariance

Proposition 3.23 (Representative invariance of the bridge). *Let $C_{\tau,W}F$ and $C'_{\tau,W}F$ be admissible realization-compatible representatives of the same windowed collapsed profile. Assume their realizations are isomorphic:*

$$\mathcal{R}(C_{\tau,W}F) \cong \mathcal{R}(C'_{\tau,W}F)$$

in $D^b(k\text{-mod})$. Then the truth value of the Ext^1 -clause is independent of the representative choice.

Proof. If the realized objects are isomorphic in $D^b(k\text{-mod})$, then their $\text{Ext}^1(-, k)$ groups are naturally isomorphic. Therefore

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0$$

if and only if

$$\text{Ext}^1(\mathcal{R}(C'_{\tau,W}F), k) = 0.$$

□

Remark 3.24 (Audit role). A certificate invoking the Ext^1 -clause must record enough representative and realization data to make Proposition 3.23 auditable. Otherwise, the Ext verdict depends on an untracked representative choice and is not Core-admissible.

3.10. Formal Core limitations of Chapter 3

The following exclusions are part of the design of this chapter.

- No theorem of the form

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \Rightarrow \text{Eval}_{1,W,\tau}(F) = 0$$

is asserted.

- No theorem of the form

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \Rightarrow \text{PH}_1(C_{\tau,W}F) = 0$$

is asserted.

- No CCC-dependent or spectral-gap-dependent reverse principle is used.
- No window-local equivalence

$$\text{PH}_1 \Leftrightarrow \text{Ext}^1$$

is stated.

- No Bridge Program or validity-map machinery is invoked.
- No problem-specific interface implication is proved in this chapter.
- No search-side potential, ranking, or AI confidence value is used to discharge the Ext^1 -clause.
- No global strict collapse functor C_τ is assumed.
- No identity

$$C_{\tau,W}F = (C_\tau F)|_W$$

is used.

- No exactness of positive-threshold collapse is used.

Remark 3.25 (Why these exclusions matter). The present chapter is the second Core nucleus of the theory. Its function is to establish exactly one direction of transport from repaired windowed topological vanishing to categorical vanishing. All stronger reverse-use mechanisms are intentionally deferred to non-Core layers so that the sovereignty of the Core remains auditable.

3.11. Provenance and status

The amplitude $[-1, 0]$ realization discipline and the Ext^1 -edge identification are the only derived-category ingredients used here. The chapter depends on Chapter 2 and Appendix A for repaired windowed collapsed persistence evaluation, and on Appendix B for admissible filtered representatives. It depends on Chapter 1 for gate semantics. It does not enlarge any of those chapters by reverse transport principles.

Status labels for this chapter. *Core mathematical theorem:* Theorem 3.13, Theorem 3.16.

Core lemma/corollary: Lemma 3.12, Corollary 3.14.

Operational / eligibility definitions: Definitions 3.2, 3.4, 3.7, 3.10, 3.18, 3.19.

Gate integration: Proposition 3.21, Proposition 3.23.

Explicit exclusion: Remark 3.1, Section 3.10.

4 Chapter 4: Failure Types, Tower Comparison Artifacts, Type IV, and Collapse Admissibility

4.1. Scope and standing regime

We work in the repaired Core regime of Chapters 1–3.

Thus:

- (i) persistence data live in constructible one-parameter persistence over a field k ;
- (ii) positive-threshold collapse is the barcode-level operation T_τ^{bar} ;
- (iii) all gate-relevant persistence data are evaluated on right-open windows $W = [u, u')$;
- (iv) the repaired windowed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F));$$

- (v) $C_{\tau,W}F$ denotes an admissible representative of $\text{Eval}_{i,W,\tau}(F)$, not a global strict collapse functor;
- (vi) Type IV diagnostics require a declared tower comparison artifact.

The purpose of this chapter is to close the failure taxonomy at the repaired Core level:

- (i) to define Types I–IV using $\text{Eval}_{i,W,\tau}$;
- (ii) to define collapse admissibility;

- (iii) to define tower comparison artifacts;
- (iv) to define the diagnostics μ_{Collapse} and u_{Collapse} ;
- (v) to formalize the monitored generic sense in which Type IV is invisible;
- (vi) to state the failure-and-diagnostics package used by B-Gate⁺.

Remark 4.1 (What is intentionally not in this chapter). This chapter does *not* develop local PH–Ext equivalence, Čech gluing, Overlap-Glue theorems, Restart/Summability propagation, or search-layer lifting protocols. Those belong either to later Core chapters or to non-Core appendices. The present chapter is only about failure taxonomy, tower comparison artifacts, and monitored Type IV diagnostics.

4.2. Failure types

Definition 4.2 (Failure types). Let $(F; W, \tau)$ be a declared windowed Core object. We distinguish the following four failure types.

Type I. Topological failure. The repaired windowed collapsed degree-1 profile is nonzero:

$$\text{Eval}_{1,W,\tau}(F) \neq 0.$$

If an admissible representative $C_{\tau,W}F$ is supplied, this may equivalently be read as

$$\text{PH}_1(C_{\tau,W}F) \neq 0.$$

Type II. Categorical failure. The triple $(F; W, \tau)$ is eligible in the sense of Chapter 3, an admissible realization-compatible representative is supplied, but

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) \neq 0.$$

Type III. Operational instability. A declared admissibility test fails to be stable under a declared finite-level pipeline operation, representative choice, windowing rule, profile comparison, or budget policy. This type records operational or functorial instability at finite level.

Type IV. Invisible tower failure. Every declared finite layer may appear admissible after windowed collapse, while a declared tower comparison artifact develops a monitored generic defect. Such defects are detected by the infinite-bar kernel/cokernel diagnostics

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}})$$

introduced below.

Remark 4.3 (Observable vs. invisible failures). Types I and II are observable on the declared window. Type III is operational. Type IV is invisible at finite level and appears only through a declared comparison between the collapsed profile of a tower and the collapsed profile of its apex or limiting object. The Core diagnostic for Type IV is the monitored generic / infinite-bar part of the kernel and cokernel of that declared comparison artifact.

4.3. Collapse admissibility

Definition 4.4 (Collapse admissibility). Let $(F; W, \tau)$ be a declared windowed Core object. We say that $(F; W, \tau)$ is *collapse-admissible* if all of the following hold:

(i) the repaired topological clause vanishes:

$$\text{Eval}_{1,W,\tau}(F) = 0;$$

(ii) if the Ext-clause is invoked, the triple is eligible in the sense of Chapter 3;

(iii) if the Ext-clause is invoked, then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0;$$

(iv) the monitored Type IV diagnostics satisfy

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0)$$

on the declared window and threshold, relative to the declared tower comparison artifact when a tower is involved.

Remark 4.5 (Relation to B-Gate⁺). Collapse admissibility is the structural part of B-Gate⁺. The full audited B-Gate⁺ of Chapter 1 also requires the budget inequality

$$\text{gap}_\tau > \Sigma\delta.$$

Thus collapse admissibility is necessary for a Core pass, but not by itself sufficient for a final audited pass.

Proposition 4.6 (One-way bridge reduces the structural burden). *Assume $(F; W, \tau)$ is eligible in the sense of Chapter 3. If*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Hence, in the eligible regime, the structural content of collapse admissibility reduces to:

(i) $\text{Eval}_{1,W,\tau}(F) = 0;$

(ii) $(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0).$

Proof. The Ext-vanishing follows from the one-way bridge of Chapter 3. The remaining conditions are exactly the topological and Type IV clauses in Definition 4.4. $\square$

4.4. Windowed collapsed tower profiles

Let

$$F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow \dots$$

be a directed tower in the declared filtered or persistence regime. Let F_∞ denote a declared apex or limiting object when such an object is available.

Definition 4.7 (Windowed collapsed tower profile). For a monitored degree i , window W , and threshold τ , the *windowed collapsed tower profile* is the profile sequence

$$\text{Eval}_{i,W,\tau}(F_0), \quad \text{Eval}_{i,W,\tau}(F_1), \quad \text{Eval}_{i,W,\tau}(F_2), \quad \dots$$

together with declared comparison data between consecutive or directed levels when such data are used.

Definition 4.8 (Collapsed apex profile). If F_∞ is a declared apex or limiting object of the tower, its collapsed apex profile is

$$\text{ApexProfile}_{i,W,\tau}(F_\infty) := \text{Eval}_{i,W,\tau}(F_\infty).$$

Definition 4.9 (Collapsed colimit profile). A collapsed colimit profile

$$\text{ColimProfile}_{i,W,\tau}(F_\bullet)$$

is declared data representing the collapsed profile obtained from the tower after applying the windowed evaluation discipline. It may be constructed from a genuine colimit in a constructible regime, from a verified stabilized profile, or from an implementation-specific profile artifact.

Remark 4.10 (No automatic colimit profile). In the repaired Core, $\text{ColimProfile}_{i,W,\tau}(F_\bullet)$ is not automatically identified with

$$\lim_{\rightarrow} T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_n))$$

unless a constructible and functorial profile construction has been declared. This is necessary because T_τ^{bar} is not a global exact functor in the general Core.

4.5. Tower comparison artifacts

Definition 4.11 (Tower comparison artifact). A *tower comparison artifact* for (i, W, τ) is declared data specifying a comparison

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

This artifact must include enough information to compute or certify the monitored kernel and cokernel profiles used by the Type IV diagnostics.

Remark 4.12 (No automatic tower comparison). The comparison

$$\phi_{i,W,\tau}$$

is not automatic in the general Core. It is either:

- (i) explicitly declared as an artifact;
- (ii) constructed in an implementation-specific profile model; or
- (iii) supplied canonically by a separately declared exact-admissible regime.

Without such data, Type IV diagnostics are not well-formed.

Definition 4.13 (Exact-admissible tower comparison). A tower comparison artifact is called *exact-admissible* if it is supplied by a regime in which:

- (i) the relevant threshold deletion is functorial on the declared tower;

- (ii) the relevant colimit or stabilized profile is constructible;
- (iii) the comparison map is canonical in that regime;
- (iv) the kernel and cokernel profiles are defined inside the monitored constructible category.

Remark 4.14 (Why the comparison is after windowed collapse). The comparison artifact is defined after the repaired windowed collapse evaluation, not before it. This matches the Core doctrine:

$$\mathbf{P}_i \longrightarrow W_W \mathbf{P}_i \longrightarrow T_\tau^{\text{bar}} \longrightarrow \text{tower comparison}.$$

Short-bar noise and window-order artifacts are therefore not silently promoted into Type IV defects.

4.6. Defect profiles and tower diagnostics

Definition 4.15 (Defect profiles). Given a tower comparison artifact

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty),$$

define the monitored defect profiles by

$$\text{Defect}_{i,W,\tau}^{\text{ker}} := \ker(\phi_{i,W,\tau}), \quad \text{Defect}_{i,W,\tau}^{\text{coker}} := \text{coker}(\phi_{i,W,\tau}),$$

whenever the declared profile regime supports the relevant kernel and cokernel computations.

Remark 4.16 (Kernel and cokernel require a declared profile regime). The expressions

$$\ker(\phi_{i,W,\tau}), \quad \text{coker}(\phi_{i,W,\tau})$$

are meaningful only inside a declared persistence-profile regime where kernels and cokernels are defined and constructible. They are not inferred merely from the existence of a tower of filtered objects.

Definition 4.17 (Generic fiber dimension). For $M \in \text{Pers}_k^{\text{cons}}$, define its *generic fiber dimension* by

$$\text{gdim}(M) := \lim_{t \rightarrow +\infty} \dim_k M(t),$$

whenever this limit is taken in the constructible regime. At the barcode level, $\text{gdim}(M)$ equals the multiplicity of bars unbounded toward $+\infty$, up to the endpoint convention fixed in Chapter 2.

Definition 4.18 (Degreewise tower diagnostics). Given defect profiles as in Definition 4.15, define

$$\mu_{i,W,\tau} := \text{gdim}(\text{Defect}_{i,W,\tau}^{\text{ker}}), \quad u_{i,W,\tau} := \text{gdim}(\text{Defect}_{i,W,\tau}^{\text{coker}}).$$

Definition 4.19 (Total collapse diagnostics). The total monitored collapse diagnostics on (W, τ) are

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad u_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} u_{i,W,\tau},$$

where I_{mon} is the finite declared set of monitored homological degrees.

Proposition 4.20 (Generic dimension and infinite-bar multiplicity). *For any morphism $\psi : M \rightarrow N$ in a constructible persistence-profile regime,*

$$\text{gdim}(\ker \psi)$$

equals the multiplicity of infinite bars in the barcode of $\ker \psi$, and likewise

$$\text{gdim}(\text{coker } \psi)$$

equals the multiplicity of infinite bars in the barcode of $\text{coker } \psi$.

Proof. In the constructible one-parameter regime, the eventual fiber dimension of a persistence module is precisely the number of interval summands unbounded toward $+\infty$. Applying this to the kernel and cokernel of ψ gives the claim. $\square$

Remark 4.21 (Interpretation of μ and u). The invariant $\mu_{i,W,\tau}$ measures persistent generic kernel defect: a monitored infinite-bar component present in the collapsed tower profile disappears at the collapsed apex profile. The invariant $u_{i,W,\tau}$ measures persistent generic cokernel defect: a monitored infinite-bar component appears in the collapsed apex profile that is not detected by the collapsed tower profile. These are the two monitored generic modes in which Type IV can occur.

Remark 4.22 (Finite-bar defects are not measured by μ , u). The diagnostics $\mu_{i,W,\tau}$ and $u_{i,W,\tau}$ measure generic / infinite-bar defects. They do not, by themselves, assert that finite-bar components of

$$\ker(\phi_{i,W,\tau}) \quad \text{or} \quad \text{coker}(\phi_{i,W,\tau})$$

vanish. Thus

$$(\mu_{i,W,\tau}, u_{i,W,\tau}) = (0, 0)$$

is weaker than saying that $\phi_{i,W,\tau}$ is an isomorphism, unless an additional no-finite-defect hypothesis is declared.

4.7. Type IV as invisible generic tower failure

Definition 4.23 (Type IV obstruction). A declared tower has a *Type IV obstruction* on (W, τ) if the tower comparison artifact exists and

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \neq (0, 0).$$

Equivalently, at least one monitored comparison artifact has a nonzero generic kernel or cokernel defect after repaired windowed collapse.

Definition 4.24 (Type IV-free in the monitored generic sense). A declared tower is said to be *Type IV-free in the monitored generic sense* on (W, τ) if its tower comparison artifact exists and

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0).$$

Theorem 4.25 (Core criterion for absence of monitored Type IV). *Relative to a declared tower comparison artifact, a tower is Type IV-free in the monitored generic sense if and only if*

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0).$$

Proof. This is precisely Definition 4.24. The diagnostics μ_{Collapse} and u_{Collapse} are the generic-dimension counts of the kernel and cokernel defects of the declared tower comparison artifacts. Their simultaneous vanishing means that no monitored infinite-bar tower defect remains. $\square$

Remark 4.26 (No automatic comparison-map isomorphism). Theorem 4.25 does not assert that every tower comparison artifact is an isomorphism. It asserts only the absence of monitored generic / infinite-bar defect. To conclude that $\phi_{i,W,\tau}$ is an isomorphism, one must also show that the finite-bar parts of

$$\ker(\phi_{i,W,\tau}) \quad \text{and} \quad \text{coker}(\phi_{i,W,\tau})$$

vanish.

Remark 4.27 (Why Type IV is called invisible). Type IV is invisible because every finite layer may pass ordinary finite-level checks after windowed collapse:

$$\text{Eval}_{1,W,\tau}(F_n) = 0, \quad \text{Ext}^1(\mathcal{R}(C_{\tau,W}F_n), k) = 0$$

when eligible, and yet the declared collapsed tower comparison may still develop a monitored generic kernel or cokernel defect. Thus finite admissibility alone does not control the declared limit or apex comparison in the monitored generic sense.

Proposition 4.28 (Finite admissibility does not imply limit admissibility). *There may exist a declared tower $\{F_n\}$, a window W , a threshold $\tau > 0$, and a tower comparison artifact such that for every finite level n ,*

$$\text{Eval}_{1,W,\tau}(F_n) = 0$$

and, whenever eligible,

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F_n), k) = 0,$$

while the declared tower comparison artifact satisfies

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \neq (0, 0).$$

In particular, finite-level admissibility need not pass to the declared apex or limit.

Proof sketch. Use a profile tower whose finite stages have no residual degree-1 bars after repaired windowed collapse, while the declared apex profile contains an infinite bar not appearing in the collapsed tower profile. The corresponding comparison artifact has a nonzero generic cokernel defect. This realizes a pure cokernel Type IV mode. $\square$

Remark 4.29 (Pure kernel, pure cokernel, mixed Type IV). Type IV admits three basic monitored generic modes:

$$(\mu > 0, u = 0), \quad (\mu = 0, u > 0), \quad (\mu > 0, u > 0).$$

The Core uses only this classification by generic comparison-artifact defect. Any geometric or problem-specific interpretation of these modes belongs to later Interface or Search layers.

4.8. Stable bands

Definition 4.30 (Stable band). Fix a window W , a monitored degree i , and a declared tower comparison policy. A *stable band* is a contiguous range

$$B \subset (0, \infty)$$

of thresholds such that, for every $\tau \in B$, the tower comparison artifact exists and

$$(\mu_{i,W,\tau}, u_{i,W,\tau}) = (0, 0).$$

Equivalently, no monitored generic Type IV defect appears in degree i anywhere on B .

Remark 4.31 (Stable band vs. comparison-map isomorphism). A stable band is a region where the monitored generic tower audit remains zero. It does not, by itself, assert that the full comparison artifacts

$$\phi_{i,W,\tau}$$

are isomorphisms on all finite-bar components. Such a stronger conclusion requires an additional no-finite-defect or full-isomorphism hypothesis.

Remark 4.32 (Why stable bands matter). A stable band is not merely a convenient region of parameter space. It is a region on which the tower audit is stable at the level of monitored Type IV detection. Thus stable bands provide the Core notion of threshold stability for generic tower diagnostics.

Proposition 4.33 (Persistence of zero diagnostics on a stable band). *If B is a stable band for degree i , then for all $\tau \in B$,*

$$(\mu_{i,W,\tau}, u_{i,W,\tau}) = (0, 0).$$

Hence no monitored generic Type IV obstruction is present on that band in degree i .

Proof. This is the defining property of a stable band. □

4.9. Interaction with B-Gate⁺

Definition 4.34 (Structural pass). A declared object $(F; W, \tau)$ has a *structural pass* if:

- (i) $\text{Eval}_{1,W,\tau}(F) = 0$;
- (ii) the Ext-clause is either not invoked or is discharged in the eligible regime;
- (iii) $(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0)$.

Proposition 4.35 (Structural pass plus budget gives B-Gate⁺). *If $(F; W, \tau)$ has a structural pass and the budget condition*

$$\text{gap}_\tau > \Sigma\delta$$

holds, then B-Gate⁺ passes on $(F; W, \tau)$.

Proof. This is the conjunction of the topological, categorical, Type IV, and budget clauses of B-Gate⁺ as fixed in Chapter 1. □

Remark 4.36 (No score replacement). No scalar score, potential Φ , advisory invariant, or Search artifact replaces the structural pass conditions or the budget condition.

4.10. Core summary of failure and admissibility

Theorem 4.37 (Failure-and-diagnostics package). *In the repaired constructible windowed-collapse Core, the following package is available.*

- (i) *The failure taxonomy splits into Types I–IV.*
- (ii) *Type I is non-vanishing of $\text{Eval}_{1,W,\tau}(F)$.*
- (iii) *Type II is non-vanishing of $\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k)$ in the eligible regime.*
- (iv) *Type III is operational or functorial instability at finite level.*
- (v) *Type IV is a monitored generic tower-level invisible failure detected by the infinite-bar kernel/cokernel defects of a declared tower comparison artifact.*
- (vi) *The invariants μ_{Collapse} and u_{Collapse} measure monitored generic kernel and cokernel defects of the declared tower comparison artifact.*
- (vii) *Collapse admissibility requires repaired topological vanishing, categorical vanishing when the Ext-clause is invoked, and absence of monitored Type IV.*

(viii) *In the eligible regime, the one-way bridge reduces the categorical burden to the repaired topological clause*

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

(ix) *Stable bands are threshold regions on which the monitored Type IV audit remains zero relative to declared comparison artifacts.*

Proof. Items (i)–(iv) follow from Definition 4.2. Items (v)–(vi) follow from Definitions 4.11, 4.15, and 4.19. Item (vii) is Definition 4.4. Item (viii) follows from Proposition 4.6. Item (ix) follows from Definition 4.30. $\square$

Remark 4.38 (Strength of the package). The package above is intentionally phrased in the monitored generic sense. It does not silently upgrade zero diagnostics into full comparison-artifact isomorphisms. This distinction is essential for keeping the Type IV audit mathematically precise under the repaired Core.

4.11. Explicit non-claims

The following are not asserted in this chapter.

(i) No automatic comparison map

$$\lim_{\xrightarrow{n}} T_{\tau}^{\text{bar}}(\mathbf{P}_i(F_n)) \rightarrow T_{\tau}^{\text{bar}}(\mathbf{P}_i(F_{\infty}))$$

is asserted in the general Core.

(ii) No automatic comparison map

$$\lim_{\xrightarrow{n}} \text{Eval}_{i,W,\tau}(F_n) \rightarrow \text{Eval}_{i,W,\tau}(F_{\infty})$$

is asserted without a declared tower comparison artifact.

(iii) No exactness of T_{τ}^{bar} is used.

(iv) No global Serre localization by positive-threshold short bars is used.

(v) No global strict filtered lift C_{τ} is used.

(vi) No identity

$$C_{\tau,W}F = (C_{\tau}F)|_W$$

is used.

(vii) No vanishing of (μ, u) is interpreted as full comparison-artifact isomorphism unless a no-finite-defect hypothesis is separately declared.

(viii) No Type IV proxy, advisory invariant, spectral warning flag, Search artifact, or AI-generated score is treated as the actual Type IV diagnostic.

(ix) No problem-specific arithmetic, PDE, geometric, or categorical interpretation of Type IV is asserted here.

4.12. Provenance and status

The material of this chapter uses the repaired windowed collapse discipline of Chapter 2 and Appendix A, the representative discipline of Appendix B, and the one-way bridge of Chapter 3. Its distinctive contribution is to place monitored generic tower failure itself inside the Core as a first-class audit object, while requiring tower comparison artifacts to be declared rather than assumed.

Status labels for this chapter. *Core definitions:* Definitions 4.2, 4.4, 4.7, 4.8, 4.9, 4.11, 4.13, 4.15, 4.17, 4.18, 4.19, 4.23, 4.24, 4.30, 4.34.

Core propositions/theorems: Proposition 4.6, Proposition 4.20, Theorem 4.25, Proposition 4.28, Proposition 4.33, Proposition 4.35, Theorem 4.37.

Operational cautions / precision remarks: Remarks 4.10, 4.12, 4.16, 4.22, 4.26, 4.31, 4.36, 4.38.

Explicit exclusions: Section 4.11.

5 Chapter 5: Window Logic, Overlap Gate, Restart/Summability, and Local-to-Global Certificate Logic

5.1. Scope and standing purpose

We continue to work in the repaired constructible windowed-collapse Core of Chapters 1–4.

The purpose of the present chapter is not to add new failure types, new bridge principles, or new Search mechanisms. Its purpose is to explain how already certified local data may be assembled into a globally coherent Core certificate record.

Accordingly, this chapter is organized around five notions:

- (i) right-open window discipline;
- (ii) Overlap Gate compatibility for repaired windowed collapsed profiles;
- (iii) finite Čech-depth control;
- (iv) Restart/Summability along MECE partitions;
- (v) countable definable-cover gluing.

Throughout this chapter, the gate-relevant local persistence datum is

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No unconditional commutation

$$W_W T_{\tau}^{\text{bar}} \cong T_{\tau}^{\text{bar}} W_W$$

is used.

Remark 5.1 (Separation from Chapter 4). Chapter 4 isolated the failure side of the Core: Types I–IV, tower comparison artifacts, monitored Type IV invisibility, and collapse admissibility. The present chapter isolates the complementary certificate side: how local repaired windowed-collapse verdicts can be compared, restarted, budgeted, and assembled into a globally coherent certificate record.

Remark 5.2 (What is intentionally excluded). This chapter does *not* use AWFS formalisms, product-ledger quantale refinements, soft-commuting policies, Mirror/Transfer 2-cell control, Window Stack machinery, or Search-layer mapping machinery. Those are intentionally excluded so that the local-to-global Core logic remains transparent and auditable.

5.2. Window logic: partitions and covers

Definition 5.3 (Right-open window). A *window* is a right-open interval

$$W = [u, u') \subset \mathbb{R}.$$

All windowed Core statements are evaluated using the repaired fixed order

$$F \mapsto \mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)) = \text{Eval}_{i,W,\tau}(F).$$

Definition 5.4 (MECE windowing). A family of right-open intervals

$$\{W_k\}_{k \in K}, \quad W_k = [u_k, u_{k+1}),$$

is called a *MECE windowing* on a declared range if:

- (i) the windows are mutually exclusive up to the fixed right-open endpoint convention;
- (ii) they are collectively exhaustive on the declared range;
- (iii) every restart step is indexed by one such declared window.

Definition 5.5 (Overlap-admitting window cover). A family of right-open windows

$$\{W_\alpha\}_{\alpha \in A}$$

is called an *overlap-admitting window cover* if the windows cover the declared range and nonempty overlaps

$$W_{\alpha_0} \cap \cdots \cap W_{\alpha_p}$$

are allowed and explicitly declared. Such covers are used for Overlap Gate and Čech-depth gluing.

Remark 5.6 (MECE partitions vs. overlapping covers). A MECE windowing and an overlap-admitting cover play different roles. MECE windowings are used for sequential restart and summability. Overlap-admitting covers are used for compatibility checks and local-to-global gluing. Conflating the two would make the Overlap Gate vacuous on strictly disjoint partitions.

Remark 5.7 (Why right-open windows are used). Right-open windows remove endpoint ambiguity in certificate accounting. They ensure that adjacent MECE windows do not duplicate endpoint data, while overlap-admitting covers may still be used when genuine overlap compatibility must be checked.

Definition 5.8 (Window-restricted certificate record). A *window-restricted certificate record* on $(F; W, \tau)$ consists of:

- (i) a declared object F ;
- (ii) the repaired windowed collapsed persistence profiles

$$\text{Eval}_{i,W,\tau}(F)$$

for the monitored degrees;

- (iii) the local B-Gate⁺ verdict or structural-pass verdict;
- (iv) the local budget data needed for overlap and restart checks;
- (v) the local monitored Type IV diagnostics, together with tower comparison artifacts if Type IV is invoked;
- (vi) representative data $C_{\tau, W}F$, only when filtered or derived clauses are used.

5.3. Overlap Gate as the local compatibility test

Definition 5.9 (Overlap budget). Let W_α and W_β be windows with nonempty overlap. The *overlap budget*

$$\Sigma\delta_{\alpha\beta}$$

is the declared budget assigned to the compatibility check on the overlap

$$W_{\alpha\beta} := W_\alpha \cap W_\beta.$$

Definition 5.10 (Overlap profile). For a monitored degree i , object F_α , overlap $W_{\alpha\beta}$, and threshold τ , define the repaired overlap profile by

$$O_{\alpha\beta, i} := \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha) = T_\tau^{\text{bar}}(\mathbb{W}_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha)).$$

Similarly,

$$O_{\beta\alpha, i} := \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta).$$

Remark 5.11 (No crop-after-collapse overlap comparison). Overlap compatibility is not computed by first collapsing on W_α and then cropping the collapsed result to $W_{\alpha\beta}$, unless a window-adapted commutation condition is declared. The default comparison is made directly on the common overlap:

$$\text{Eval}_{i, W_{\alpha\beta}, \tau}(-).$$

This avoids using the generally false commutation

$$\mathbb{W}_{W_{\alpha\beta}} T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} \mathbb{W}_{W_{\alpha\beta}}.$$

Definition 5.12 (Overlap Gate). Let W_α and W_β be overlapping windows. The *Overlap Gate* passes on $W_{\alpha\beta} = W_\alpha \cap W_\beta$ if the following hold.

- (i) **Overlap discrepancy is controlled.** The repaired overlap profiles agree up to the declared overlap budget:

$$d_{\text{int}}(\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha), \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta)) \leq \Sigma\delta_{\alpha\beta}$$

for all monitored degrees i .

- (ii) **Budget margin is positive.** The safety margin dominates the overlap budget:

$$\text{gap}_\tau > \Sigma\delta_{\alpha\beta}.$$

- (iii) **No monitored generic Type IV appears on the overlap.** The tower diagnostics satisfy

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0)$$

on the declared overlap and monitored degrees. If Type IV diagnostics are invoked, the relevant overlap tower comparison artifact must be declared.

Remark 5.13 (Meaning of the Overlap Gate). The Overlap Gate is the Core compatibility test for local certificate records. It does not create a certificate by itself. Rather, it certifies that two already declared local certificates do not conflict beyond the admitted budget and do not hide a monitored generic Type IV defect on the overlap.

5.4. Čech depth and finite overlap complexity

Definition 5.14 (Finite Čech depth). An overlap-admitting cover $\{W_\alpha\}_{\alpha \in A}$ has *finite Čech depth* K if no nonempty overlap of order $K + 1$ occurs:

$$W_{\alpha_0} \cap \cdots \cap W_{\alpha_K} = \emptyset$$

for all distinct $\alpha_0, \dots, \alpha_K$. Equivalently, the Čech nerve of the cover has dimension at most $K - 1$.

Lemma 5.15 (Finite overlap reduction). *If a window cover has finite Čech depth K , then all overlap compatibility checks needed for Core local-to-global gluing reduce to overlap levels of order at most K .*

Proof. By definition, no overlap of order $K + 1$ is nonempty. Hence the Čech nerve terminates in degree $K - 1$, so no higher-order compatibility data are required. $\square$

Remark 5.16 (Why Čech depth belongs to the Core). Finite Čech depth is not a search convenience. It is a Core finiteness condition ensuring that gluing remains auditable and that local certificate propagation can be checked by finitely many overlap strata.

5.5. Local-to-global certificate logic

Definition 5.17 (Local certificate family). A *local certificate family* on an overlap-admitting cover $\{W_\alpha\}$ at threshold τ is a family of window-restricted certificate records such that each member passes the local structural conditions of collapse admissibility, together with the declared local budget and diagnostic data.

Definition 5.18 (Globally coherent certificate family). A local certificate family is *globally coherent* if:

- (i) the Overlap Gate passes on every nonempty declared overlap;
- (ii) the cover has finite Čech depth;
- (iii) all budget and diagnostic data required by the overlap checks are declared in the certificate record;
- (iv) all Type IV checks used in the overlaps have declared tower comparison artifacts.

Theorem 5.19 (Local-to-global certificate principle). *Let $\{W_\alpha\}$ be a declared right-open overlap-admitting cover of finite Čech depth. Assume:*

- (i) *each local window W_α carries a window-restricted certificate record;*
- (ii) *each local record is collapse-admissible on its own window;*
- (iii) *the Overlap Gate passes on every nonempty declared overlap;*
- (iv) *all required budgets, diagnostics, representative data, and comparison artifacts are declared.*

Then the family determines a globally coherent Core certificate record on the union

$$\bigcup_{\alpha} W_{\alpha}.$$

Proof. Finite Čech depth reduces the compatibility problem to finitely many overlap levels by Lemma 5.15. The Overlap Gate guarantees agreement of repaired windowed collapsed profiles within the admitted budget and excludes monitored generic Type IV defects on overlaps. Hence the local certificate records can be assembled into a globally coherent Core certificate record. $\square$

Remark 5.20 (Certificate record, not object-level gluing). Theorem 5.19 is a theorem about certificate records and repaired windowed collapsed profile compatibility. It does not assert an unrestricted sheaf-theoretic, filtered-chain, or object-level gluing theorem. Any stronger object-level gluing statement must be separately declared and proved.

Remark 5.21 (Why this is a Core theorem). This theorem is about certificate propagation, not about heuristic map-building. Its role is to justify when the Core may replace a family of local passes by a globally coherent certificate record on the covered region. For this reason it belongs to the Core and not to the Search layer.

5.6. Restart lemma on MECE partitions

Definition 5.22 (Per-window pipeline budget). Let $\{W_k\}$ be a MECE windowing with thresholds τ_k . For each window W_k , define the total declared budget

$$\Sigma\delta_k$$

as the sum of all admitted algorithmic, discretization, measurement, representative, and comparison contributions used before the local B-Gate⁺ check on W_k .

Lemma 5.23 (Restart lemma). *Assume B-Gate⁺ passes on a window W_k with margin*

$$\text{gap}_{\tau_k} > \Sigma\delta_k.$$

Suppose the next window W_{k+1} is reached through declared admissible continuation steps, followed by repaired windowed collapse at threshold τ_{k+1} . Then there exists a constant $\kappa \in (0, 1]$, depending only on the declared continuation class and threshold policy, such that

$$\text{gap}_{\tau_{k+1}} \geq \kappa(\text{gap}_{\tau_k} - \Sigma\delta_k).$$

Proof. By the declared continuation policy, admissible continuation may decrease the available margin only in a controlled manner. The factor κ records the worst allowed degradation compatible with the continuation class and threshold policy. The repaired windowed collapse operation is non-expansive at the barcode-profile level, so it does not amplify the declared continuation error. $\square$

Remark 5.24 (Meaning of restart). The Restart lemma says that a local certificate may be propagated forward only when margin is not exhausted by the already accumulated budget. Thus restart is not free. It is a controlled inheritance principle for certified windows in a MECE sequence.

5.7. Summability

Definition 5.25 (Summability policy). A certified run satisfies *summability* if, for the monitored MECE windows,

$$\sum_k \Sigma\delta_k < \infty$$

in the declared budget semantics.

Remark 5.26 (Why summability is necessary). Without summability, window-by-window restarting can leak an unbounded amount of budget. In that case, even if every individual local check appears acceptable, the global certificate record ceases to be auditable.

Theorem 5.27 (Restart–Summability principle). *Let $\{W_k\}$ be a MECE windowing. Assume:*

- (i) $B\text{-Gate}^+$ passes on each W_k ;
- (ii) the Restart lemma applies at every transition $W_k \rightarrow W_{k+1}$;
- (iii) the run satisfies summability;
- (iv) all representative and tower comparison data used in each window are declared.

Then the windowwise local certificate records assemble into a globally coherent Core certificate record on

$$\bigcup_k W_k.$$

Proof. The Restart lemma ensures that positive margin can be propagated from one certified window to the next under the declared continuation policy. Summability guarantees that the total accumulated budget remains finite. The declared representative and comparison data keep the windowwise verdicts auditable. Hence the propagated local certificate records remain globally coherent and can be assembled into a globally coherent Core certificate record. $\square$

Remark 5.28 (Restart versus overlap gluing). Restart/Summability and Overlap Gate gluing are complementary. Restart/Summability controls sequential inheritance along a MECE partition. Overlap Gate gluing controls compatibility on overlapping covers. Both produce globally coherent certificate records, but by different mechanisms.

5.8. Countable definable covers

Definition 5.29 (Countable definable cover). A *countable definable cover* is a countable family of right-open windows

$$\{W_n\}_{n \geq 1}$$

such that:

- (i) each W_n is definable in the declared ambient structure;
- (ii) the family covers the declared range;
- (iii) the cover has finite Čech depth.

Theorem 5.30 (Countable definable-cover gluing). Let $\{W_n\}_{n \geq 1}$ be a countable definable cover of finite Čech depth. Assume:

- (i) each W_n carries a window-restricted certificate record;
- (ii) the Overlap Gate passes on every nonempty declared overlap;
- (iii) the corresponding overlap and pipeline budgets are summable;
- (iv) all required diagnostics, representative data, tower comparison artifacts, and artifact references are declared.

Then the local certificate records assemble into a globally coherent Core certificate record on the covered range.

Proof. Finite Čech depth reduces the gluing problem to finitely many overlap levels. Summability prevents divergence of the cumulative budget. The Overlap Gate guarantees compatibility on every declared overlap and excludes monitored generic Type IV defects on those overlaps. Declared representative and tower comparison artifacts ensure that local structural verdicts are reproducible. Therefore the local certificate records assemble into a globally coherent Core certificate record. $\square$

Remark 5.31 (Relation to Chapter 1). This theorem is the local-to-global counterpart of the finite-window logic already implicit in Chapter 1. The present chapter makes explicit that countable coverage is allowed in the Core only when definability, finite Čech depth, summability, and artifact-backed auditability keep the overlap structure controlled.

5.9. Core summary of window logic and gluing

Theorem 5.32 (Window-and-gluing package). *Within the repaired windowed-collapse Core, the following package is available.*

- (i) *Right-open windows provide the admissible local domains of certification.*
- (ii) *MECE windowings govern sequential restart and summability.*
- (iii) *Overlap-admitting covers govern compatibility and local-to-global gluing.*
- (iv) *The default local persistence datum is*

$$\text{Eval}_{i,w,\tau}(F) = T_{\tau}^{\text{bar}}(\mathbb{W}_w \mathbf{P}_i(F)).$$

- (v) *The Overlap Gate compares repaired overlap profiles directly on the common overlap.*
- (vi) *Finite Čech depth bounds the complexity of overlap verification.*
- (vii) *The Restart lemma controls certificate inheritance across adjacent certified MECE windows.*
- (viii) *Summability controls cumulative budget growth.*
- (ix) *Local collapse-admissible certificate records may be assembled into a globally coherent Core certificate record when overlap compatibility, finite Čech depth, and summability are all present.*
- (x) *Countable definable covers are admissible in the Core exactly under these gluing conditions.*

Remark 5.33 (Strength of this package). The package above is deliberately formulated at the level of certificate records and repaired windowed collapsed profiles. It does not assert unrestricted object-level gluing, nor does it import Search-layer map construction into the Core.

5.10. Explicit non-claims

The following are not asserted in this chapter.

- (i) No object-level sheaf gluing theorem is asserted.
- (ii) No filtered-chain gluing theorem is asserted.

(iii) No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is used.

(iv) No overlap comparison is performed by cropping already-collapsed local profiles unless a window-adapted commutation condition is declared.

(v) No global exactness of T_τ^{bar} is assumed.

(vi) No global strict C_τ is assumed.

(vii) No Type IV diagnostic is accepted without a declared tower comparison artifact.

(viii) No Search-layer validity map is treated as a globally coherent Core certificate.

(ix) No AI-generated map, score, confidence, or proposal replaces Overlap Gate, Restart/Summability, or B-Gate⁺.

5.11. Provenance and status

This chapter uses the repaired window discipline of Chapter 1, the fixed windowed collapsed evaluation order of Chapter 2, the eligible one-way bridge discipline of Chapter 3, and the monitored Type IV diagnostics of Chapter 4. Its function is to explain how certified local records become globally coherent certified records.

Status labels for this chapter. *Core definitions:* Definitions [5.3](#), [5.4](#), [5.5](#), [5.8](#), [5.9](#), [5.10](#), [5.12](#), [5.14](#), [5.17](#), [5.18](#), [5.22](#), [5.25](#), [5.29](#).

Core lemmas/theorems: Lemma [5.15](#), Theorem [5.19](#), Lemma [5.23](#), Theorem [5.27](#), Theorem [5.30](#), Theorem [5.32](#).

Operational cautions / precision remarks: Remarks [5.6](#), [5.11](#), [5.20](#), [5.28](#), [5.33](#).

Explicit exclusion: Section 5.10.

6 Chapter 6: δ -ledger, gap_τ , Budget Semantics, Audit Discipline, and Core Sovereignty

6.1. Scope and purpose

The purpose of this chapter is to isolate the audit semantics of the repaired AK Core. The present chapter does not introduce new persistence objects, new bridges, new tower diagnostics, or new local-to-global gluing principles. Instead, it specifies how already-defined structural data are converted into auditable accept/reject verdicts.

All audit statements in this chapter are read relative to the repaired windowed collapsed evaluation

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

If a filtered or derived clause is invoked, it must use an admissible representative

$$C_{\tau,W}F$$

of the relevant windowed collapsed profile.

More precisely, this chapter makes the following items explicit:

- (i) the meaning of the δ -ledger;
- (ii) the role of the safety margin gap_τ ;
- (iii) the budget inequality

$$\Sigma\delta < \text{gap}_\tau;$$

- (iv) the Core accept/reject discipline;
- (v) the sovereignty of the Core over all Search-side quantities.

Remark 6.1 (What is intentionally excluded). This chapter does *not* develop geometric collapse programs, normalization heuristics, higher comparison architectures, Search algorithms, terrain maps, lifting strategies, or AI-agent protocols. Those belong to technical extensions or to the Search/Platform layer, not to the audit Core.

6.2. The δ -ledger

Definition 6.2 (δ -ledger). The δ -*ledger* is the declared budget object recording all admissible deviations between the ideal repaired windowed-collapse pipeline and the implemented, discretized, represented, compared, or measured one. It takes values in the commutative quantale fixed in Chapter 1.

Definition 6.3 (Ledger components). The total ledger contribution on a declared window is decomposed into admissible components, typically of the form

$$\Sigma\delta = \delta^{\text{alg}} \oplus \delta^{\text{disc}} \oplus \delta^{\text{meas}} \oplus \delta^{\text{rep}} \oplus \delta^{\text{cmp}},$$

where:

- (i) δ^{alg} records algorithmic or implementation-level discrepancy;
- (ii) δ^{disc} records discretization or sampling discrepancy;
- (iii) δ^{meas} records measurement or observation-level discrepancy;
- (iv) δ^{rep} records representative-choice discrepancy, when $C_{\tau,W}F$ is used;
- (v) δ^{cmp} records comparison-artifact discrepancy, when tower or overlap comparison artifacts are used.

Additional declared components are allowed, provided they are explicitly recorded and aggregated by the stated quantale law.

Definition 6.4 (Total budget). For a declared window W , threshold τ , and monitored degree i , the total budget

$$\Sigma\delta(W, \tau, i)$$

is the quantale-sum of all ledger components relevant to the audited verdict on (W, τ, i) . When the context is fixed, we write simply

$$\Sigma\delta.$$

Remark 6.5 (Ledger semantics). The δ -ledger is not an informal error bar. It is part of the certification object. Every admissible verdict in the Core is relative to a declared ledger semantics and a declared aggregation rule.

Remark 6.6 (Ledger after v18.1 repair). Because the repaired Core uses windowed collapsed profiles, representatives, and tower comparison artifacts, the ledger must record not only numerical or discretization errors but also any declared representative and comparison tolerances. A proof object that omits these contributions is not audit-complete.

6.3. Safety margin and budget inequality

Definition 6.7 (Safety margin). For a declared threshold τ , the quantity

$$\text{gap}_\tau > 0$$

is the *safety margin* available for audited certification at that threshold. It is the admissible slack below which the local Core verdict remains stable under the declared budget semantics.

Definition 6.8 (Budget pass condition). A declared windowed run satisfies the *budget pass condition* if

$$\Sigma\delta < \text{gap}_\tau.$$

Remark 6.9 (Strict inequality). The inequality is strict. The boundary case

$$\Sigma\delta = \text{gap}_\tau$$

is not treated as a pass in the Core. This convention prevents boundary cases from being silently reclassified as certified.

Proposition 6.10 (Monotonic meaning of the budget condition). *If a declared certificate satisfies*

$$\Sigma\delta < \text{gap}_\tau,$$

then any certificate obtained by replacing $\Sigma\delta$ with a smaller declared total budget remains budget-admissible. Conversely, once

$$\Sigma\delta \geq \text{gap}_\tau,$$

the Core budget condition fails.

Proof. This is immediate from the order structure of the declared budget semantics. □

6.4. Budget semantics as part of the Core

Definition 6.11 (Budget semantics). A *budget semantics* consists of:

- (i) the quantale in which ledger values live;
- (ii) the aggregation law $\oplus$;
- (iii) the interpretation of each component of $\Sigma\delta$;
- (iv) the interpretation of $\Sigma\delta$ on a declared run;
- (v) the threshold-dependent safety margin gap_τ ;
- (vi) the strict pass/fail rule comparing $\Sigma\delta$ with gap_τ .

Remark 6.12 (Why budget semantics belongs to the Core). Without a declared budget semantics, the phrases “within tolerance” or “small enough” would be external to the theorem-bearing framework. The AK audit discipline instead requires that admissible budget comparison be part of the Core itself.

6.5. Audit discipline and Core verdicts

Definition 6.13 (Audit pass). Let $(F; W, \tau)$ be a declared windowed object. The *Audit Mode* passes on $(F; W, \tau)$ if and only if B-Gate⁺ passes on $(F; W, \tau)$ in the sense of Chapter 1. Equivalently, unpacking B-Gate⁺, all of the following hold:

- (i) the repaired topological gate passes:

$$\text{Eval}_{1,W,\tau}(F) = 0;$$

- (ii) if the Ext-clause is invoked, an admissible realization-compatible representative $C_{\tau,W}F$ is supplied and

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0;$$

- (iii) the monitored generic Type IV diagnostics vanish:

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0);$$

- (iv) if Type IV diagnostics are invoked, the required tower comparison artifact is declared and auditable;

- (v) the budget pass condition holds:

$$\Sigma\delta < \text{gap}_{\tau}.$$

Definition 6.14 (Audit reject). The *Audit Mode* rejects on $(F; W, \tau)$ if B-Gate⁺ fails on $(F; W, \tau)$. Equivalently, rejection occurs if at least one of the following holds:

- (i) the repaired topological gate fails:

$$\text{Eval}_{1,W,\tau}(F) \neq 0;$$

- (ii) the Ext-clause is invoked but no admissible realization-compatible representative is supplied;

- (iii) the Ext-clause is invoked in the eligible regime and

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) \neq 0;$$

- (iv) the monitored generic Type IV diagnostics are nonzero:

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \neq (0, 0);$$

- (v) Type IV diagnostics are invoked but the required tower comparison artifact is missing;

- (vi) the budget condition fails:

$$\Sigma\delta \geq \text{gap}_{\tau}.$$

Theorem 6.15 (Audit discipline). *In the Core, acceptance is determined only by:*

- (i) the repaired windowed collapsed profile verdicts;
- (ii) eligible Ext-clause verdicts when invoked;
- (iii) monitored generic tower diagnostics;
- (iv) declared tower comparison artifacts when Type IV is invoked;

(v) the budget comparison

$$\Sigma\delta < \text{gap}_\tau.$$

No other quantity is part of the final audited pass/reject rule.

Proof. This is a direct restatement of Definitions 6.13 and 6.14. $\square$

Remark 6.16 (Accept/reject is not heuristic). The audit verdict is a structural and budgeted verdict. It is not a confidence score, not a ranking, and not a plausibility estimate. Any softer interpretation belongs outside the Core.

Remark 6.17 (Relation to collapse admissibility). Collapse admissibility from Chapter 4 is the structural part of the audit verdict. Audit pass additionally requires the budget condition

$$\Sigma\delta < \text{gap}_\tau.$$

Thus structural admissibility and audited acceptance are related but not identical.

6.6. Core sovereignty

Definition 6.18 (Core sovereignty). The *Core sovereignty principle* is the rule that only Core-certified objects and Core audit conditions may determine a Core verdict. Equivalently, no external proposal mechanism may overwrite the output of the Core audit discipline.

Theorem 6.19 (Search quantities do not update Core verdicts). *Let Q_{search} be any Search-side quantity, including but not limited to a potential Φ , a terrain score, a ranking, a candidate priority, a validity-map value, an AI-generated confidence estimate, or an agent-generated proof sketch. Then Q_{search} does not alter the Core pass/reject verdict on a declared window.*

Proof. By Definition 6.18, a Core verdict is determined only by Core-certified objects and the audit discipline of Theorem 6.15. A Search-side quantity is not one of these. Therefore it has no authority to update the verdict. $\square$

Remark 6.20 (Meaning of Φ). The quantity Φ , when present, is only a navigation quantity. It may guide where the Search layer proposes to look next. It does not certify, discharge, repair, or override any Core gate condition.

Remark 6.21 (Core sovereignty after the repair). Even the repaired collapse operation itself contributes to a Core verdict only through declared windowed evaluation records

$$\text{Eval}_{i,w,\tau}(F)$$

and, when needed, admissible representative and tower comparison artifacts. No informal statement that an object has “collapsed” is a Core verdict.

6.7. Proposer–verifier separation

Definition 6.22 (Proposer–verifier separation). A *proposer* is any mechanism producing candidate windows, candidate thresholds, candidate objects, candidate trajectories, candidate representatives, candidate comparison artifacts, or candidate interface hypotheses for further audit. A *verifier* is the Core audit discipline acting through repaired windowed evaluations, gates, monitored diagnostics, declared artifacts, and budget checks.

Proposition 6.23 (Separation principle). *Every Search-side mechanism is a proposer. The Core is the verifier. No proposer output becomes a theorem, certificate, or pass verdict until it is re-entered into the verifier discipline of the Core and satisfies the required evaluation, gate, diagnostic, artifact, and budget conditions.*

Proof. This is a direct consequence of Theorem 6.19 and the definitions of Audit pass/reject. $\square$

Remark 6.24 (Why this matters structurally). Once Search is separated from verification in this way, the Search layer can be modified, extended, or even replaced without changing the theorem-bearing Core. This is exactly why Core sovereignty must be stated in the main text.

6.8. Minimal manifest consequences

Proposition 6.25 (Audit-readable manifest data). *A proof object is audit-readable only if it exposes, at minimum:*

- (i) *the declared windows W ;*
- (ii) *the declared threshold τ ;*
- (iii) *the collapse policy:*

$$\text{barcode_level}, \quad \text{tau_split}, \quad \text{exact_admissible};$$
- (iv) *the repaired evaluation records*

$$\text{Eval}_{i,W,\tau}(F);$$
- (v) *the representative policy and representative artifacts, if $C_{\tau,W}F$ is used;*
- (vi) *the declared ledger policy;*
- (vii) *the declared value of $\Sigma\delta$;*
- (viii) *the declared value of gap_{τ} ;*
- (ix) *the gate verdicts;*
- (x) *the relevant monitored tower diagnostics;*
- (xi) *the tower comparison artifacts, if Type IV diagnostics are invoked;*
- (xii) *the artifact references needed to reconstruct these data.*

Proof. These are exactly the data required by the audit pass/reject definitions together with the Minimal Manifest Axiom of Chapter 1. Without them, the audit verdict cannot be reconstructed. $\square$

Remark 6.26 (Relation to Chapter 1). Chapter 1 introduced the Minimal Manifest Axiom. The present chapter clarifies why budget data, evaluation records, representative data, and comparison artifacts are indispensable parts of that axiom: they are not auxiliary metadata but part of the logical input to the verifier.

6.9. Failure of auditability

Definition 6.27 (Auditability failure). A declared run has an *auditability failure* if a Core verdict is asserted while at least one of the required manifest components in Proposition 6.25 is missing.

Proposition 6.28 (Auditability failure implies no Core verdict). *If a run has an auditability failure, then it cannot be assigned a Core pass verdict.*

Proof. A Core pass verdict requires reconstructible verification of the repaired evaluation, gate, diagnostic, artifact, and budget conditions. If required manifest components are missing, the verification cannot be reconstructed. Therefore the run cannot be assigned a Core pass verdict. $\square$

Remark 6.29 (Distinction from mathematical failure). An auditability failure is not necessarily a mathematical counterexample. It means that the run has not supplied enough data to be treated as a certified proof object. This distinction prevents missing artifacts from being silently interpreted as mathematical success.

6.10. Summary package

Theorem 6.30 (Audit-and-sovereignty package). *Within the repaired Core, the following package is available.*

- (i) *The δ -ledger records admissible budgeted deviation.*
- (ii) *The ledger may include algorithmic, discretization, measurement, representative, and comparison-artifact components.*
- (iii) *The safety margin gap_τ determines the threshold for budget admissibility.*
- (iv) *The budget condition is*

$$\Sigma\delta < \text{gap}_\tau.$$
- (v) *Audit pass/reject is determined only by repaired windowed collapsed profile verdicts, eligible Ext verdicts when invoked, monitored generic tower diagnostics, declared artifacts, and the budget condition.*
- (vi) *Search-side quantities, including Φ , validity-map values, and AI confidence scores, do not update Core verdicts.*
- (vii) *The Core is the verifier; Search is only a proposer.*
- (viii) *Auditability is itself a Core requirement: missing manifest data prevents a Core pass verdict.*
- (ix) *This proposer–verifier separation is part of the theorem-bearing discipline of the AK Core.*

Proof. Items (i)–(iv) follow from Definitions 6.2, 6.3, 6.7, and 6.8. Item (v) is Theorem 6.15. Items (vi)–(vii) follow from Theorem 6.19 and Proposition 6.23. Item (viii) follows from Proposition 6.28. Item (ix) summarizes the Core sovereignty discipline. $\square$

6.11. Explicit non-claims

The following are not asserted in this chapter.

- (i) No Search-side score certifies a Core verdict.
- (ii) No scalar potential Φ replaces B-Gate⁺.
- (iii) No AI-generated confidence value replaces an audit record.
- (iv) No missing representative artifact is treated as harmless when the Ext-clause is invoked.
- (v) No missing tower comparison artifact is treated as harmless when Type IV diagnostics are invoked.
- (vi) No equality

$$\Sigma\delta = \text{gap}_\tau$$

is treated as a pass.

- (vii) No informal statement that a profile has “collapsed” replaces the declared record

$$\text{Eval}_{i,w,\tau}(F).$$

- (viii) No global exactness of T_τ^{bar} is assumed.
- (ix) No global strict filtered lift C_τ is assumed.

6.12. Provenance and status

This chapter contains no geometric extension material. Its role is normative and audit-theoretic: to state how structural data become audited verdicts, and to protect the theorem-bearing Core from contamination by Search-side proposal mechanisms.

Status labels for this chapter. *Core definitions:* Definitions [6.2](#), [6.3](#), [6.4](#), [6.7](#), [6.8](#), [6.11](#), [6.13](#), [6.14](#), [6.18](#), [6.22](#), [6.27](#).

Core propositions/theorems: Proposition [6.10](#), Theorem [6.15](#), Theorem [6.19](#), Proposition [6.23](#), Proposition [6.25](#), Proposition [6.28](#), Theorem [6.30](#).

Operational cautions / precision remarks: Remarks [6.16](#), [6.17](#), [6.20](#), [6.21](#), [6.24](#), [6.26](#), [6.29](#).

Explicit exclusion: Remark [6.1](#), Section 6.11.

7 Chapter 7: Core Theorem Package and Explicit Non-Claims

7.1. Scope and purpose

The purpose of this chapter is to close the theorem-bearing Core of the present text after the v18.1 repair. Chapters 1–6 introduced the guard-rails, the repaired barcode-level collapse discipline, the representative discipline, the one-way bridge, the failure diagnostics, the local-to-global certificate logic, and the audit discipline.

The present chapter records, in a single place:

- (i) what the repaired Core proves as mathematics;
- (ii) what the Core fixes as operational audit discipline;
- (iii) what is explicitly not claimed.

Remark 7.1 (Why this chapter is necessary). The strength of the AK framework does not lie only in what it proves. It also lies in the discipline with which it refuses to overstate what has not been proved. For this reason, explicit non-claims are part of the architecture rather than editorial afterthoughts.

Remark 7.2 (Mathematical theorems vs. operational discipline). Not every Core-level statement is a mathematical theorem about external objects. Some Core-level statements are operational audit axioms, manifest requirements, artifact requirements, or reading rules. This chapter therefore separates the mathematical package, the operational package, and the non-claim package.

7.2. The repaired Core mathematical package

Definition 7.3 (Repaired Core mathematical package). The *repaired Core mathematical package* is the collection of mathematical theorems, lemmas, and propositions established in Chapters 1–6 under the standing guard-rails of constructible one-parameter persistence over a field, repaired windowed collapse evaluation, admissible representative discipline, and declared tower comparison artifacts.

Theorem 7.4 (Contents of the repaired Core mathematical package). *The following statements belong to the repaired Core mathematical package.*

- (i) **Constructible persistence regime.** *The Core works in the declared constructible one-parameter persistence category*

$$\text{Pers}_k^{\text{cons}}$$

over a field k , with barcode statements confined to that regime.

- (ii) **Barcode-level threshold collapse.** *For each threshold $\tau > 0$, the operation*

$$T_\tau^{\text{bar}}$$

deletes finite bars of length at most τ . It is endpoint-independent, idempotent, threshold-monotone, and non-expansive at the barcode metric level.

- (iii) **No general positive-threshold Serre localization.** *For $\tau > 0$, short-bar objects are not treated as a Serre subcategory in general. Therefore T_τ^{bar} is not asserted to be a global exact Serre reflector or abelian localization on all of*

$$\text{Pers}_k^{\text{cons}}.$$

- (iv) **Windowed collapsed evaluation.** *For a right-open window W , the Core windowed persistence profile is evaluated by the fixed order*

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is not asserted.

(v) **Admissible filtered representatives.** A filtered representative

$$C_{\tau, W}F$$

is used only when filtered or derived-level constructions are required. It must satisfy

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F)$$

for monitored degrees under the declared representative equivalence policy. It is not a global strict collapse functor by default.

(vi) **Eligible realization and Ext-edge.** In the declared eligible regime, realized representatives lie in amplitude

$$D^{[-1, 0]}(k\text{-mod}),$$

and for

$$A \in D^{[-1, 0]}(k\text{-mod})$$

one has

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

(vii) **One-way bridge.** In the eligible regime, the repaired windowed topological vanishing

$$\text{Eval}_{1, W, \tau}(F) = 0$$

equivalently, for an admissible representative,

$$\text{PH}_1(C_{\tau, W}F) = 0,$$

implies

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

(viii) **Failure taxonomy.** The Core distinguishes Types I–IV. Type I is non-vanishing of $\text{Eval}_{1, W, \tau}(F)$. Type II is Ext failure in the eligible regime. Type III is operational or functorial instability. Type IV is monitored generic tower-level invisible failure.

(ix) **Tower comparison artifacts.** Type IV diagnostics are defined relative to declared tower comparison artifacts

$$\phi_{i, W, \tau} : \text{ColimProfile}_{i, W, \tau}(F_{\bullet}) \longrightarrow \text{ApexProfile}_{i, W, \tau}(F_{\infty}).$$

Such artifacts are not automatic in the general Core.

(x) **Tower diagnostics.** The diagnostics

$$\mu_{\text{Collapse}}, \quad u_{\text{Collapse}}$$

are defined from the generic fiber dimensions of the kernel and cokernel defect profiles of declared tower comparison artifacts. Their vanishing means absence of monitored generic Type IV defect. It does not automatically imply that every comparison artifact is an isomorphism on all finite-bar components.

(xi) **Collapse admissibility.** Collapse admissibility requires:

$$\text{Eval}_{1, W, \tau}(F) = 0,$$

categorical vanishing in the eligible regime when the Ext-clause is invoked, and absence of monitored generic Type IV defect.

(xii) **Local-to-global certificate logic.** *Right-open windows, overlap-admitting covers, Overlap Gate, finite Čech-depth control, Restart/Summability, and countable definable-cover gluing provide a Core mechanism for assembling local certificate records into globally coherent certificate records.*

(xiii) **Audit and budget discipline.** *A structural pass becomes an audited pass only when the budget condition*

$$\Sigma\delta < \text{gap}_\tau$$

also holds.

Remark 7.5 (Strength of the mathematical package). The mathematical package is intentionally phrased at the level proved in Chapters 1–6. In particular, monitored generic Type IV vanishing is not silently upgraded to full comparison-artifact isomorphism, and certificate-record gluing is not silently upgraded to unrestricted object-level sheaf gluing.

7.3. The repaired Core operational package

Definition 7.6 (Core operational package). The *Core operational package* is the collection of audit rules, manifest requirements, gate protocols, artifact requirements, and reading disciplines that govern admissible certification inside the Core. These rules are binding on admissible runs, but they are not themselves theorems about external mathematical objects.

Proposition 7.7 (Contents of the Core operational package). *The following statements belong to the Core operational package.*

- (i) **Claim taxonomy.** *Every statement must be read as a Core theorem, Core lemma, operational policy, interface theorem, Search artifact, or non-claim.*
- (ii) **Minimal Manifest Axiom.** *Every certified proof object must expose declared windows, threshold policy, collapse policy, evaluation order, representative policy when used, ledger policy, diagnostics, gate verdicts, tower comparison artifacts when Type IV is invoked, and artifact references.*
- (iii) **Collapse policy.** *The certified run must declare whether it uses*

$$\text{barcode_level}, \quad \text{tau_split}, \quad \text{exact_admissible}$$

collapse status.

- (iv) **B-Gate⁺.** *The Core gate passes only when the repaired topological gate, eligible categorical gate when invoked, monitored generic Type IV diagnostics, declared artifact requirements, and budget condition all pass.*
- (v) **δ -ledger.** *All admissible deviations are recorded in a declared δ -ledger with declared budget semantics and declared aggregation law. The ledger may include algorithmic, discretization, measurement, representative, and comparison-artifact components.*
- (vi) **Budget condition.** *The strict inequality*

$$\Sigma\delta < \text{gap}_\tau$$

is required for audited acceptance. The boundary case

$$\Sigma\delta = \text{gap}_\tau$$

is not a pass.

(vii) **Audit discipline.** Core pass/reject is determined only by:

- (a) repaired windowed collapsed profile verdicts;
- (b) eligible Ext verdicts when invoked;
- (c) monitored generic tower diagnostics;
- (d) declared tower comparison artifacts when Type IV is invoked;
- (e) the budget comparison

$$\Sigma\delta < \text{gap}_\tau.$$

(viii) **Auditability requirement.** Missing required manifest data prevents a Core pass verdict.

(ix) **Core sovereignty.** Search-side quantities do not update Core verdicts.

(x) **Proposer-verifier separation.** Search is a proposer. The Core audit discipline is the verifier. No proposal becomes a certificate without re-entry through the Core verifier.

Remark 7.8 (Why operational discipline is still Core-level). The operational package is not theorem-bearing in the same sense as the Ext-edge theorem or the one-way bridge. Nevertheless, it is part of the Core because it determines what counts as an admissible proof object, an admissible run, and an admissible verdict.

7.4. Status classes beyond the Core

Definition 7.9 (Status classes beyond the Core). Statements appearing in the larger AK document may fall into any of the following non-Core or conditionally Core-adjacent classes.

- (i) **Operational policy.** A certification rule, manifest rule, audit convention, or protocol governing admissible runs.
- (ii) **Interface theorem.** A theorem that combines Core material with explicitly declared problem-specific hypotheses.
- (iii) **Technical extension.** A mathematically meaningful extension that is not required for the shortest theorem-bearing Core.
- (iv) **Search artifact.** A proposal object, map, ranking, trajectory, terrain report, or navigation output generated outside the verifier discipline.
- (v) **Platform artifact.** A run log, proof-store entry, schema, agent output, replay trace, or execution record used to support audit but not itself a theorem.
- (vi) **[Spec] statement.** A declared specification, design target, external realization schema, or conditional claim not established as a Core theorem.

Remark 7.10 (Interpretation rule for [Spec]). A statement labeled [Spec] may be useful, coherent, and even strategically central, but it is not part of the Core mathematical package unless it is explicitly promoted and reproved under the Core guard-rails.

7.5. Formal non-claims

Definition 7.11 (Explicit non-claims). The following statements are *not* asserted by the Core.

Theorem 7.12 (Explicit non-claims of the Core). *The Core does not claim any of the following.*

(i) **No global reverse bridge.** *The Core does not claim*

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0 \implies \mathrm{Eval}_{1, W, \tau}(F) = 0$$

in general.

(ii) **No global PH–Ext equivalence.** *The Core does not claim a global equivalence*

$$\mathrm{PH}_1 \Leftrightarrow \mathrm{Ext}^1.$$

(iii) **No automatic transport from local reverse-use hypotheses.** *Any CCC-dependent, spectral-gap-dependent, or otherwise locally authorized reverse-use principle is not part of the Core unless reproved inside it.*

(iv) **No positive-threshold Serre subcategory claim.** *The Core does not claim that the class of finite bars of length at most $\tau > 0$ forms a Serre subcategory in general.*

(v) **No global exact reflector.** *The Core does not claim that*

$$T_{\tau}^{\mathrm{bar}}$$

is a global exact reflector, exact endofunctor, abelian localization, or Serre localization on all of

$$\mathrm{Pers}_k^{\mathrm{cons}}.$$

(vi) **No quotient formula in the general Core.** *The Core does not claim a general formula*

$$T_{\tau}(M) = M/E_{\tau}(M)$$

for positive-threshold collapse unless a separately declared exact-admissible regime justifies it.

(vii) **No unconditional crop–collapse commutation.** *The Core does not claim*

$$\mathbb{W}_W T_{\tau}^{\mathrm{bar}} \cong T_{\tau}^{\mathrm{bar}} \mathbb{W}_W$$

without a declared τ -window-adapted or exact-admissible condition.

(viii) **No global strict filtered lift.** *The Core does not claim a global strict functor*

$$C_{\tau} : \mathrm{FiltCh}((\cdot)k) \rightarrow \mathrm{FiltCh}((\cdot)k)$$

in the general regime.

(ix) **No default restriction identity.** *The Core does not claim*

$$C_{\tau, W}F = (C_{\tau}F)|_W$$

without a declared window-compatibility condition.

(x) **No automatic tower comparison.** The Core does not claim an automatic comparison map

$$\lim_{\substack{\longrightarrow \\ n}} \text{Eval}_{i,w,\tau}(F_n) \rightarrow \text{Eval}_{i,w,\tau}(F_\infty)$$

without a declared tower comparison artifact or exact-admissible regime.

(xi) **No full-isomorphism conclusion from zero μ , u .** The Core does not claim that

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) = (0, 0)$$

implies full comparison-artifact isomorphism on all finite-bar components. It means absence of monitored generic / infinite-bar Type IV defect.

(xii) **No unrestricted object-level gluing.** The Core local-to-global theorem glues certificate records. It does not assert an unrestricted sheaf-theoretic, filtered-chain, or object-level gluing theorem.

(xiii) **No theorem status for validity maps.** A Map of Validity is not a Core theorem.

(xiv) **No theorem status for Search-layer reports.** A Search-layer report is not a theorem.

(xv) **No theorem status for AI-generated outputs.** An AI agent's ranking, confidence, proposed correction, heuristic prioritization, or generated proof sketch is not a reason for Core acceptance.

(xvi) **No promotion from proposal to proof.** A candidate object, candidate region, candidate threshold, candidate representative, candidate comparison artifact, candidate interface, or candidate trajectory does not become certified merely by being found.

(xvii) **No unconditional domain resolution claims.** The Core does not claim to solve any external problem family merely because a realization scheme, calibration protocol, or search procedure has been described.

(xviii) **No unstated interface transport.** Problem-specific consequences are not claimed without an explicitly declared interface hypothesis and a clearly stated transport theorem.

(xix) **No theorem status for implementation convenience.** A numerically convenient, visually convincing, or computationally stable output is not thereby a Core certificate.

(xx) **No replacement of gate logic by scalar scores.** No scalar quantity, including a potential Φ , replaces or weakens gate verdicts, monitored diagnostics, declared artifact requirements, or budget checks.

(xxi) **No hidden extension of the Core by appendices.** Material placed in technical, interface, search, platform, or companion appendices does not enlarge the Core theorem package unless explicitly imported and reproofed.

Remark 7.13 (Interpretive consequence). Whenever a reader encounters a stronger statement elsewhere in the project, the burden is on that statement to declare its status. If it is not explicitly proved inside the Core regime, it must be read as operational, interface-level, technical, search-side, platform-side, companion-side, or [Spec].

7.6. Boundary between theorem, policy, and artifact

Definition 7.14 (Theorem/policy/artifact boundary). The following boundary rules are binding.

- (i) A *theorem* asserts mathematically stable content under the declared Core guard-rails.
- (ii) A *policy* governs admissible certification procedure but is not itself a theorem about external mathematical objects.
- (iii) An *artifact* is evidence, output, trace, report, representative data, comparison data, or generated object from a run, but not a theorem.

Proposition 7.15 (Boundary discipline). *No policy is to be read as a theorem. No artifact is to be read as a theorem. No artifact becomes a theorem without re-entry through the theorem-bearing verifier discipline.*

Proof. This follows from the Claim Taxonomy of Chapter 1, the audit discipline of Chapter 6, and the explicit non-claims of Theorem 7.12. $\square$

Remark 7.16 (Why this is more than wording). The theorem/policy/artifact boundary is structural. Without it, the document would permit silent inflation from operational success to mathematical proof. The present chapter forbids such inflation.

7.7. Canonical reading rules for later chapters and appendices

Definition 7.17 (Canonical reading rules). Later chapters, appendices, platforms, and companions are to be read under the following rules.

- (i) If a statement depends only on Chapters 1–6 and is proved under their repaired guard-rails, it may belong to the Core mathematical package.
- (ii) If a statement governs admissible certification but is not a theorem about mathematical objects, it belongs to the operational package.
- (iii) If a statement depends on problem-specific hypotheses, it is an interface statement unless reproved in the Core.
- (iv) If a statement depends on Search protocols, validity maps, AI agents, bridge programs, or platform orchestration, it is not a Core theorem.
- (v) If a statement depends on exactness of positive-threshold collapse, it must declare an exact-admissible regime.
- (vi) If a statement depends on $C_{\tau, W}F = (C_{\tau}F)|_W$, it must declare a window-compatibility condition.
- (vii) If a statement depends on tower diagnostics, it must declare the tower comparison artifact.
- (viii) If a statement is marked [Spec], it is not theorem-bearing Core content.
- (ix) If status is ambiguous, the weaker classification prevails.

Remark 7.18 (Default conservatism). The AK reading rule is conservative by design. Whenever doubt exists about whether a statement has theorem status, it must be treated as non-Core until the stronger status is explicitly justified.

7.8. The Core closure principle

Theorem 7.19 (Core closure principle). *The repaired Core mathematical package is closed under explicit reuse of theorems, definitions, and guard-rails already established in Chapters 1–6, but it is not enlarged by:*

- (i) *heuristic search success;*
- (ii) *empirical map-building;*
- (iii) *implementation convenience;*
- (iv) *platform-side orchestration;*
- (v) *AI-agent confidence;*
- (vi) *undeclared transport across interfaces;*
- (vii) *undeclared exactness of positive-threshold collapse;*
- (viii) *undeclared tower comparison artifacts;*
- (ix) *undeclared representative choices;*
- (x) *appendical or companion material not explicitly imported and reproved.*

Proof. Each excluded enlargement mechanism is explicitly ruled out either by the audit discipline of Chapter 6 or by the non-claim discipline of Theorem 7.12. $\square$

Remark 7.20 (Why this closes the main body). This theorem is the closure statement for the main theorem-bearing body. It says that the Core expands only by proof and declared admissible artifacts, not by surrounding architecture. That is the point of ending the Core body with explicit non-claims.

7.9. Summary package

Theorem 7.21 (Core package and non-claim package). *The main body of the repaired AK Core closes with the following three packages.*

- (i) **Core mathematical package.** *Constructible persistence, barcode-level threshold collapse, repaired windowed collapsed evaluation, admissible representatives, eligible realization, Ext-edge identification, the one-way bridge, monitored generic Type IV diagnostics via declared tower comparison artifacts, collapse admissibility, and local-to-global certificate-record logic.*
- (ii) **Core operational package.** *Claim taxonomy, Minimal Manifest Axiom, collapse policy, representative policy, B-Gate⁺, δ -ledger, budget semantics, audit pass/reject discipline, Core sovereignty, auditability failure, and proposer–verifier separation.*
- (iii) **Non-claim package.** *No global reverse bridge, no global $PH_1 \Leftrightarrow Ext^1$, no positive-threshold Serre localization, no global exact reflector T_τ^{bar} , no unconditional crop–collapse commutation, no global strict C_τ , no automatic tower comparison, no full-isomorphism conclusion from zero μ , u , no unrestricted object-level gluing, no theorem status for validity maps, no theorem status for Search reports, no theorem status for AI confidence, no scalar-score replacement for gates, and no hidden extension of the Core by appendical, platform, or companion material.*

7.10. Provenance and status

This chapter contains no new persistence calculation, no new derived-category argument, and no new tower invariant. Its role is juridical in the mathematical sense: to state exactly what has been proved, what is operationally required, what has not been claimed, and how later material must be read.

Status labels for this chapter. *Core definitions:* Definitions 7.3, 7.6, 7.9, 7.11, 7.14, 7.17.

Core theorems/propositions: Theorem 7.4, Proposition 7.7, Theorem 7.12, Proposition 7.15, Theorem 7.19, Theorem 7.21.

Interpretive doctrine: Remarks 7.1, 7.2, 7.5, 7.8, 7.10, 7.13, 7.16, 7.18, 7.20.

8 Chapter 8: Interface Protocols, High-Dimensional Projection, and Closure of the Main Body

8.1. Scope and purpose

The purpose of this chapter is to close the main body of the repaired AK Core by specifying how the Core may be connected to external problem domains without enlarging the Core mathematical package.

The central rule of this chapter is:

Projection creates Core-readable input; it does not create Core verdicts.

This chapter does not solve any arithmetic, PDE, geometric, normalization, categorical, analytic, symplectic, or computational problem. It does not present calibration examples, case studies, numerical experiments, Search outcomes, validity maps, or AI-generated reports. It defines only the protocol by which such domains may be projected into Core-readable form and then evaluated through the already established repaired Core verifier.

Remark 8.1 (Protocol rather than application). The present chapter is not an application chapter. Its role is to define admissible interfaces. Any domain-specific conclusion drawn through such an interface is conditional on the declared hypotheses of that interface and does not retroactively upgrade the Core.

Remark 8.2 (Relation to Chapter 7). Chapter 7 closed the Core mathematical, operational, and non-claim packages. The present chapter explains how later appendices, companions, Search layers, platform layers, and problem-interface documents may connect external domains to the Core without altering those packages.

Remark 8.3 (Why this chapter is needed). Without an explicit interface discipline, the following invalid inference would be tempting:

external structure is successfully represented $\implies$ external theorem is proved.

This chapter forbids that inference. Representation, projection, realization, calibration, Search success, empirical stability, or platform execution may prepare objects for Core verification. They do not by themselves establish Core pass, Core reject, or any external mathematical theorem.

8.2. High-dimensional projection as the AK interface mechanism

The original AK viewpoint is that high-dimensional structures are not judged directly. They are examined through projected, realized, or represented shadows that live in the Core-readable persistence or filtered regime.

In the repaired v18.1 discipline, this viewpoint is retained, but its logical location is made explicit:

High-dimensional projection belongs to the Interface mechanism.

Collapse-based degeneration judgment belongs to the Core verifier mechanism.

Definition 8.4 (External source class). An *external source class* is any declared family of objects outside the minimal Core persistence regime. Examples include:

- (i) arithmetic source data, such as Galois representations, Selmer complexes, Iwasawa modules, congruence data, or L -function profiles;
- (ii) PDE source data, such as initial data, solution trajectories, vorticity fields, energy profiles, or discretized flow observables;
- (iii) geometric or analytic source data, such as moduli points, degenerating families, singularity models, spectral data, or normalization problems;
- (iv) categorical source data, such as derived categories, sheaf categories, Fukaya categories, A_∞ -structures, or comparison functors;
- (v) computational or Search-side source data, such as candidate cells, lifted objects, validity maps, platform traces, or proof candidates.

An external source class is not a Core object by default.

Definition 8.5 (AK high-dimensional projection datum). An *AK high-dimensional projection datum* consists of the following data:

- (i) an external source class $\mathcal{D}$;
- (ii) an external object $X \in \mathcal{D}$;
- (iii) a declared projection, representation, extraction, or realization procedure

$$\mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{FiltCh}(\cdot)k \quad \text{or} \quad \mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{Pers}_k^{\text{cons}};$$

- (iv) the projected object

$$F_X := \mathfrak{R}_{AK}(X);$$

- (v) declared right-open windows $W = [u, u')$, thresholds $\tau > 0$, and monitored degrees i ;
- (vi) the repaired windowed evaluation policy

$$\text{Eval}_{i,W,\tau}(F_X) = T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F_X));$$

- (vii) a declared ledger policy recording all projection, realization, measurement, discretization, comparison, approximation, normalization, or transport defects;

(viii) artifact references sufficient for reconstruction and audit.

The datum is interface-level data unless its realization properties are separately proved under the Core guard-rails.

Remark 8.6 (Projection is structural, not merely linear). The word “projection” in AK theory is structural. It does not mean only a linear projection, coordinate projection, dimensional reduction, embedding, or numerical compression. It means any declared procedure by which features of a high-dimensional or external object become readable as filtered or persistence data.

Thus an AK projection may be geometric, arithmetic, analytic, categorical, spectral, computational, functorial, tower-theoretic, or hybrid, provided that its target, purpose, and defect ledger are explicitly declared.

Definition 8.7 (Core-readable projected object). Let $X \in \mathcal{D}$ and $F_X = \mathfrak{R}_{AK}(X)$. The object F_X is *Core-readable* on (W, τ, i) if the repaired windowed collapsed profile

$$\text{Eval}_{i,W,\tau}(F_X) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_X))$$

is defined in the constructible persistence regime, and all required representative data, eligibility data, diagnostics, ledger entries, tower comparison artifacts, and artifact references are declared.

Definition 8.8 (Projection-to-collapse evaluation protocol). For every admissible interface, the AK evaluation order is:

$$X \mapsto F_X = \mathfrak{R}_{AK}(X) \mapsto \mathbf{P}_i(F_X) \mapsto W_W \mathbf{P}_i(F_X) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_X)) = \text{Eval}_{i,W,\tau}(F_X) \mapsto \text{Core gate and diagnostics.}$$

Equivalently, the Core never judges the raw external object X directly. It judges only the repaired windowed collapsed persistence data obtained from the declared projection.

Proposition 8.9 (Projection does not determine Core verdict). *An AK high-dimensional projection datum does not by itself determine Core pass or Core reject.*

Proof. A projection datum supplies a typed route from an external object to a Core-readable filtered or persistence object. Core pass/reject is determined only after the repaired windowed evaluation, gate checks, monitored Type IV diagnostics, ledger comparison, and manifest verification. Therefore projection provides input to the Core verifier, but it does not replace the Core verifier. $\square$

Remark 8.10 (Preservation of the original AK viewpoint). This formulation preserves the original AK intuition:

high-dimensional structure $\rightsquigarrow$ projected persistence shadow $\rightsquigarrow$ collapse-based degeneration judgment.

What v18.1 changes is not this intuition, but its logical placement. Projection is an Interface-side mechanism. Degeneration judgment is a Core verifier-side mechanism.

8.3. Interface protocol principles

Definition 8.11 (Interface protocol principles). Every admissible interface from an external problem family into the AK Core must satisfy the following principles.

- (i) **Declared source and projection.** The external source class, the object under study, and the projection or realization procedure must be declared.

- (ii) **Purpose-faithful realization.** There must be a declared hypothesis specifying how objects from the external domain are realized as filtered or persistence objects in a way that preserves the features needed for the stated Core test.
- (iii) **Declared Core-readable target.** The target of the interface must be explicitly typed as a filtered chain object, a constructible persistence object, a repaired windowed collapsed profile, an eligible realized object, or non-Core advisory data.
- (iv) **Constructibility.** Any persistence object used for Core certification must lie in the constructible one-parameter persistence regime over the coefficient field k .
- (v) **Eligible regime.** Any use of the Ext^1 -clause must pass through the eligible regime of Chapter 3.
- (vi) **Repaired windowed collapse only.** All Core comparisons, gates, diagnostics, certificate records, and gluing statements must be evaluated through

$$\text{Eval}_{i,w,\tau}(F) = T_{\tau}^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)).$$

- (vii) **Problem-specific implication is [Spec] by default.** Any implication from external-domain structure to Core-visible behavior, or from Core-visible behavior to an external-domain conclusion, is specification-level unless separately promoted and proved.
- (viii) **Pre-gate/Core-gate separation.** Domain-specific screening, preprocessing, calibration, normalization, sanity checks, or Search filters may exist as pre-gates, but they do not replace, redefine, or weaken the Core gate.
- (ix) **Manifest discipline.** Any certificate produced through an interface must carry the declared windows, thresholds, collapse policy, ledger policy, diagnostics, gate verdicts, eligibility data when needed, tower comparison artifacts when needed, and artifact references required for audit.

Remark 8.12 (Minimality of the protocol). These principles intentionally do not prescribe a specific arithmetic realization, PDE observable, spectral construction, normalization scheme, categorical comparison functor, or Search strategy. They prescribe only the admissible shape of the connection to the Core.

8.4. Purpose-faithful projection and realization

Definition 8.13 (Purpose-faithful realization hypothesis). A *purpose-faithful realization hypothesis* is a declared statement of the form

$$\mathfrak{R}_{\text{dom}} : \mathcal{D}_{\text{ext}} \longrightarrow \text{FiltCh}(()k) \quad \text{or} \quad \mathfrak{R}_{\text{dom}} : \mathcal{D}_{\text{ext}} \longrightarrow \text{Pers}_k^{\text{cons}},$$

where $\mathcal{D}_{\text{ext}}$ denotes the declared external domain, together with the claim that the chosen realization preserves the features needed for the intended Core test.

Remark 8.14 (Purpose-faithful is not categorical faithfulness). The phrase “purpose-faithful” is operational. It does not mean that $\mathfrak{R}_{\text{dom}}$ is faithful in the strongest categorical sense. It means faithful for the declared interface purpose.

For example, if an interface is designed to test a Selmer-like obstruction, it must preserve the relevant obstruction profile. It need not preserve every arithmetic invariant of the original object. If a PDE interface is designed to test high-vorticity concentration, it must preserve the declared persistence features of that observable. It need not preserve every analytic invariant of the flow.

Definition 8.15 (Declared realization target). An interface must declare whether its target is one of the following:

- (i) a filtered chain object in $\text{FiltCh}(()k)$;
- (ii) a constructible persistence object in $\text{Pers}_k^{\text{cons}}$;
- (iii) a repaired windowed collapsed profile $\text{Eval}_{i,w,\tau}(F)$;
- (iv) an admissible representative $C_{\tau,w}F$;
- (v) a realized eligible object in $D^{[-1,0]}(k\text{-mod})$;
- (vi) a ledger-compatible certificate component;
- (vii) a tower comparison artifact;
- (viii) a non-Core advisory or Search artifact.

Definition 8.16 (Interface purpose declaration). An *interface purpose declaration* records:

- (i) the external phenomenon being tested;
- (ii) the Core-visible feature intended to represent it;
- (iii) the monitored degree or degrees;
- (iv) the window and threshold policy;
- (v) the admissible defect sources;
- (vi) whether the intended output is an interface pass, a Core pass, an advisory signal, a bridge candidate, or an external-domain theorem candidate.

Proposition 8.17 (No interface without declared realization). *No problem-specific connection to the Core is admissible unless a purpose-faithful realization hypothesis and a declared realization target have been specified.*

Proof. Without a declared realization, the external domain is not connected to the persistence-level or filtered-level objects on which the Core operates. Without a declared target, the relevant Core gate, bridge, diagnostic, or certificate-record condition has no typed input. Hence no admissible interface exists. $\square$

Remark 8.18 (Realization can fail). A realization hypothesis may fail. It may fail because the target is not constructible, because the filtration is not one-parameter, because the intended observable is not purpose-faithful, because the ledger cannot bound the transfer defect, because eligibility is unavailable, or because required artifacts are missing. Failure of realization is an interface failure. It is not automatically a Core reject.

8.5. Core-readable interface objects

Definition 8.19 (Core-readable interface object). Let X be an external object and let

$$F_X = \mathfrak{R}_{AK}(X).$$

The object F_X is a *Core-readable interface object* if, for every declared window W , threshold τ , and monitored degree i , the expression

$$\text{Eval}_{i,W,\tau}(F_X) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_X))$$

is defined in the constructible persistence regime, and the interface supplies all required representative, eligibility, diagnostic, ledger, comparison, and artifact data.

Definition 8.20 (Interface certificate input). An *interface certificate input* consists of:

- (i) the external source object X ;
- (ii) the projection or realization route $\mathfrak{R}_{AK}$;
- (iii) the Core-readable object F_X ;
- (iv) declared windows W ;
- (v) threshold policy τ , or a declared τ -sweep;
- (vi) monitored degrees;
- (vii) admissible representative data when required;
- (viii) eligible realization data when the Ext^1 -clause is invoked;
- (ix) tower comparison artifacts when Type IV diagnostics are invoked;
- (x) ledger entries for all transfer, approximation, comparison, measurement, discretization, normalization, or transport defects;
- (xi) artifact references.

Proposition 8.21 (Constructibility is mandatory for Core readability). *If a projected persistence object is wild, nonconstructible, infinite-rank without control, or lacks locally finite critical behavior on monitored windows, then it is not Core-readable.*

Proof. The Core operates in constructible one-parameter persistence over a field. Its repaired windowed evaluation, barcode-level threshold deletion, diagnostics, and gate conditions are stated in that regime. A projected object outside that regime has no typed access to the Core verifier unless it is first replaced by a declared constructible approximation with ledgered error or otherwise re-entered through a proved admissible route. $\square$

Remark 8.22 (Approximation is not concealment). If an external object is approximated, discretized, sampled, normalized, or compressed in order to become Core-readable, the approximation defect must be declared in the ledger. An approximation may support a certificate only through declared error control and the budget condition. It may not be hidden inside the word “projection.”

8.6. Eligibility transfer requirement

Definition 8.23 (Eligibility transfer requirement). Let $\mathfrak{R}_{\text{dom}}$ be a purpose-faithful realization hypothesis. An interface satisfies the *eligibility transfer requirement* if, whenever the interface invokes the Ext^1 -side of the Core, the realized representative belongs to the eligible regime of Chapter 3.

Concretely, for the windowed representative $C_{\tau, W}F$, one must have

$$\mathcal{R}(C_{\tau, W}F) \in D^{[-1, 0]}(k\text{-mod})$$

together with the edge-realization condition required for the Ext^1 -edge.

Remark 8.24 (Why eligibility is non-negotiable). Eligibility is not an optional refinement. Without it, the Ext^1 -edge identification of the Core is not available. Therefore any interface theorem using Ext^1 without declared eligibility lies outside the Core protocol.

Proposition 8.25 (No categorical discharge without eligibility). *If an interface does not satisfy the eligibility transfer requirement, then the Ext^1 -clause of the Core may not be invoked through that interface.*

Proof. The Ext^1 -edge and the one-way bridge are available only under the eligible realization regime. External origin does not weaken this requirement. Hence an interface lacking eligibility cannot use the Ext^1 -clause for Core discharge. $\square$

Remark 8.26 (External Ext is not automatically Core Ext). Arithmetic Ext-groups, deformation Ext-groups, sheaf Ext-groups, categorical Ext-groups, or obstruction groups appearing in an external theory are not automatically the Core group

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k).$$

Any identification must be declared as an interface hypothesis or proved as an interface theorem.

8.7. Repaired after-collapse interface discipline

Definition 8.27 (After-collapse interface discipline). An interface satisfies the *after-collapse interface discipline* if every Core-relevant comparison, gate evaluation, diagnostic computation, certificate-record construction, and gluing step is carried out through the repaired windowed evaluation

$$\text{Eval}_{i, W, \tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Any deviation from this order must be explicitly justified and ledgered.

Remark 8.28 (Why interfaces inherit the repaired after-collapse doctrine). The after-collapse doctrine is not confined to abstract persistence theory. It is inherited by every admissible interface. Otherwise the interface would measure a different object from the one on which the Core theorems are stated.

Proposition 8.29 (No raw-domain substitution for collapsed Core objects). *A domain-specific observable, invariant, simulation output, numerical field, arithmetic datum, categorical object, or Search artifact cannot be substituted directly for a collapsed Core object unless it is first placed inside the declared purpose-faithful realization and then evaluated through $\text{Eval}_{i, W, \tau}$.*

Proof. The Core theorems apply to repaired windowed collapsed objects in the declared persistence/filtered regime. A raw-domain object that bypasses this path is not of the right type for direct Core use. $\square$

Definition 8.30 (Interface-side evaluation order). An admissible interface must evaluate Core-relevant data in the following order:

$$\begin{aligned}
X &\mapsto F_X = \mathfrak{R}_{AK}(X) \\
&\mapsto \mathbf{P}_i(F_X) \\
&\mapsto W_W \mathbf{P}_i(F_X) \\
&\mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_X)) \\
&= \text{Eval}_{i,W,\tau}(F_X) \\
&\mapsto B\text{-Gate}^+ \text{ and Type IV diagnostics.}
\end{aligned}$$

Equivalently,

$$X \mapsto F_X \mapsto \text{Eval}_{i,W,\tau}(F_X) \mapsto B\text{-Gate}^+ \text{ and Type IV diagnostics.}$$

8.8. Collapse-based degeneration judgment

Definition 8.31 (Interface degeneration judgment). An *interface degeneration judgment* is a Core-side judgment about the repaired windowed collapsed profile of the projected object F_X . It is not a direct judgment about the raw external object X .

On a declared (W, τ, i) , the basic forms are:

- (i) topological degeneration, detected by residual degree-1 persistence:

$$\text{Eval}_{1,W,\tau}(F_X) \text{ eq } 0;$$

- (ii) categorical degeneration, detected by residual Ext^1 in the eligible regime:

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F_X), k) \text{ eq } 0;$$

- (iii) tower-level invisible degeneration, detected by

$$(\mu_{\text{Collapse}}, u_{\text{Collapse}}) \text{ eq } (0, 0);$$

- (iv) budget-level inadmissibility, detected by failure of

$$\Sigma\delta < \text{gap}_\tau.$$

Definition 8.32 (Projected degeneration profile). The *projected degeneration profile* of X under $\mathfrak{R}_{AK}$ is the collection of Core-visible gate verdicts, Type I–IV diagnostics, and ledger comparisons obtained from

$$\text{Eval}_{i,W,\tau}(\mathfrak{R}_{AK}(X))$$

over the declared windows, thresholds, and monitored degrees.

Proposition 8.33 (Degeneration judgment is Core-side). A *projected degeneration profile* may support a Core verdict for the projected object, but it does not by itself establish an external-domain theorem about X .

Proof. The projected degeneration profile concerns the Core-readable object $\mathfrak{R}_{AK}(X)$ after repaired windowed collapse. An external-domain theorem concerns X in its original semantic setting. A transport theorem is needed to connect the Core-visible profile back to the external-domain claim. $\square$

Remark 8.34 (Pure high-dimensional projection reading). In the original AK reading, degeneration is detected by examining whether the projected persistence shadow collapses cleanly. In the v18.1 discipline, this is formalized as follows:

$$X \text{ is not judged directly; } \quad \text{Eval}_{i,W,\tau}(\mathfrak{R}_{AK}(X)) \text{ is judged by the Core.}$$

This is the repaired v18.1 form of high-dimensional projection-based degeneration analysis.

8.9. Problem-specific transport and default $[Spec]$ status

Definition 8.35 (Problem-specific implication). A *problem-specific implication* is any statement of the form domain-specific property or failure $\implies$ Core-visible persistence, realization, diagnostic, or certificate behavior, or conversely,

$$\text{Core certificate behavior} \implies \text{domain-specific conclusion.}$$

Definition 8.36 (Interface-to-Core transport statement). An *interface-to-Core transport statement* is a problem-specific implication of the form

$$\text{external-domain structure} \implies \text{Core-visible behavior.}$$

Definition 8.37 (Core-to-interface transport statement). A *Core-to-interface transport statement* is a problem-specific implication of the form

$$\text{Core certificate behavior} \implies \text{external-domain conclusion.}$$

Definition 8.38 (Interface transport theorem). An *interface transport theorem* is a transport statement supplied with:

- (i) a precise external source class;
- (ii) a purpose-faithful projection or realization hypothesis;
- (iii) a declared Core-readable target;
- (iv) constructibility data;
- (v) admissible representative data when representatives are used;
- (vi) eligibility data if Ext^1 is used;
- (vii) tower comparison artifacts if Type IV diagnostics are used;
- (viii) all required ledgered transfer or comparison defects;
- (ix) artifact references;
- (x) a proof that the stated external-domain and Core-side conditions correspond.

Definition 8.39 (Default status of problem-specific implications). Every problem-specific implication is classified as $[Spec]$ by default. It is not part of the Core mathematical package unless separately promoted and reproved under the Core guard-rails.

Remark 8.40 (Why the default is necessary). The Core is a structural theorem package about repaired windowed collapsed profiles, admissible representatives, eligible realization, the one-way bridge, monitored Type IV diagnostics, certificate records, and audit. The moment a statement imports arithmetic meaning, PDE meaning, normalization meaning, symplectic meaning, categorical comparison meaning, or platform meaning, it leaves the pure Core unless the imported content is explicitly internalized and reproved.

Theorem 8.41 (Interface transport does not enlarge the Core by default). *Let an interface connect an external domain to the Core via a purpose-faithful projection or realization hypothesis. Then any problem-specific implication used by that interface remains $[Spec]$ unless explicitly promoted and reproved as a Core theorem or as a separately declared interface theorem.*

Proof. By the Claim Taxonomy, Core theorems are those proved inside the standing Core regime. A problem-specific implication uses external semantics in addition to the Core regime. Therefore it remains interface-level or $[Spec]$ unless the additional semantic content is explicitly imported and reproved under the Core guard-rails. $\square$

Remark 8.42 (Transport directions are distinct). The directions

$$\text{external-domain structure} \implies \text{Core behavior}$$

and

$$\text{Core behavior} \implies \text{external-domain conclusion}$$

are logically distinct. Neither direction is automatic. An interface may prove one direction, both directions, or neither.

8.10. Pre-gates, advisory quantities, and Core gates

Definition 8.43 (Pre-gate). A *pre-gate* is any domain-specific screening or preprocessing device used before Core certification. Examples include:

- (i) arithmetic local-condition checks;
- (ii) conductor, rank, reduction, or congruence prechecks;
- (iii) PDE regularity, stability, resolution, or solver sanity checks;
- (iv) filtration-orientation checks;
- (v) discretization and measurement checks;
- (vi) normalization prerequisites;
- (vii) categorical comparison sanity checks;
- (viii) Search-side priority filters;
- (ix) AI-generated candidate rankings.

Definition 8.44 (Core gate). The *Core gate* is the gate discipline already fixed by the AK Core, in particular:

$$B\text{-Gate}^+, \quad (\mu_{\text{Collapse}}, u_{\text{Collapse}}), \quad \Sigma\delta < \text{gap}_\tau.$$

Definition 8.45 (Advisory interface quantity). An *advisory interface quantity* is any scalar, profile, score, map, warning flag, spectral indicator, numerical residual, Search potential, or AI-generated confidence value used to guide inspection, but not to determine Core pass/reject.

Proposition 8.46 (Pre-gate/Core-gate separation). A *pre-gate* may restrict which domain objects are submitted to the Core, but it may not:

- (i) *redefine* $B\text{-Gate}^+$;
- (ii) *weaken monitored Type IV diagnostics*;
- (iii) *modify the budget inequality*;

- (iv) *replace the eligible regime;*
- (v) *convert an advisory quantity into a gate verdict without explicit budget conversion and Core re-entry;*
- (vi) *upgrade a failing Core verdict into a passing one.*

Proof. Pre-gates operate before Core certification. Core pass/reject is determined by verifier-side Core data: repaired windowed evaluation records, gate verdicts, monitored diagnostics, declared artifacts, and budget comparison. Therefore a pre-gate can filter inputs but cannot replace the Core verifier. $\square$

Remark 8.47 (Pre-gate failure versus Core reject). Failure of a pre-gate means that the object is not suitable for the declared interface route. It is not automatically a Core reject. Conversely, passing all pre-gates does not imply Core acceptance.

8.11. Interface certificate records and manifest discipline

Definition 8.48 (Interface manifest entry). An *interface manifest entry* records:

- (i) the external source object;
- (ii) the external source class;
- (iii) the projection or realization route;
- (iv) the purpose-faithful realization hypothesis;
- (v) the declared realization target;
- (vi) declared windows and thresholds;
- (vii) monitored degrees;
- (viii) constructibility evidence;
- (ix) admissible representative data when representatives are used;
- (x) eligibility data if Ext^1 is invoked;
- (xi) tower comparison artifacts if Type IV diagnostics are invoked;
- (xii) transfer, comparison, approximation, discretization, measurement, normalization, or transport defects;
- (xiii) ledger entries and aggregation policy;
- (xiv) Core gate verdicts;
- (xv) monitored Type IV diagnostics;
- (xvi) artifact references;
- (xvii) status labels for all problem-specific implications;
- (xviii) any advisory quantities, clearly marked as non-verdict data.

Definition 8.49 (Interface certificate record). An *interface certificate record* is a certificate record whose source data are external and whose Core-readable data are obtained through a declared interface manifest entry.

Proposition 8.50 (Interface certificate records must satisfy Minimal Manifest). *Every interface certificate record must satisfy the Minimal Manifest Axiom.*

Proof. External source data do not exempt a certificate from Core auditability. The Minimal Manifest Axiom requires declared windows, threshold policy, collapse policy, ledger policy, diagnostics, gate verdicts, artifact references, and comparison artifacts when needed. Therefore an interface certificate record must satisfy the same minimum requirements. $\square$

Remark 8.51 (Artifacts support reconstruction, not verdicts). Artifacts may include source equations, datasets, simulation logs, barcodes, Galois data, Selmer complexes, tower layers, solver outputs, realization scripts, formal proof traces, categorical comparison data, or platform logs. They support reconstruction. They do not by themselves determine Core pass.

Definition 8.52 (Interface pass). An *interface pass* means that the external source data have been successfully projected or realized into Core-readable form under the declared interface protocol.

Definition 8.53 (Core pass for interface data). A *Core pass for interface data* means that the corresponding Core certificate record passes the Core verifier, including B-Gate⁺, monitored Type IV diagnostics, declared artifact requirements, and the budget condition.

Definition 8.54 (External-domain conclusion). An *external-domain conclusion* is a problem-specific statement about the original external object, such as an arithmetic theorem, PDE regularity statement, geometric normalization statement, categorical equivalence, or symplectic conclusion.

Proposition 8.55 (Core pass is not external-domain conclusion). *A Core pass for interface data does not imply an external-domain conclusion unless a separate transport theorem proves that implication.*

Proof. A Core pass certifies Core-visible persistence, diagnostic, artifact, and budget conditions. An external-domain conclusion concerns the semantics of the original external object. A theorem connecting the two is required. Without such a theorem, the implication is not valid. $\square$

Remark 8.56 (Three-layer distinction). The following must remain distinct:

interface pass, Core pass, external-domain theorem.

Confusing these layers violates the non-claim discipline of Chapter 7.

8.12. Admissible interface families

Definition 8.57 (Arithmetic interface family). An *arithmetic interface family* is a declared class of arithmetic source data together with a proposed realization route into filtered objects, constructible persistence objects, eligible realized objects, tower comparison artifacts, or certificate-record components.

Arithmetic interfaces are admissible only under the general protocol of this chapter. In particular, arithmetic realization is an interface, not an arithmetic theorem.

Definition 8.58 (PDE interface family). A *PDE interface family* is a declared class of PDE source data, observables, solutions, approximations, or numerical fields together with a proposed route into Core-readable filtered or persistence data.

PDE interfaces are admissible only when measurement, smoothing, discretization, solver, and approximation defects are declared and ledgered.

Definition 8.59 (Geometric or normalization interface family). A *geometric or normalization interface family* is a declared class of geometric objects, singularity models, moduli data, normalization procedures, or degeneration families together with a proposed persistence or filtered realization.

Such interfaces may propose Core-visible obstruction profiles, but they do not prove geometric conclusions without transport theorems.

Definition 8.60 (Categorical or higher-structural interface family). A *categorical or higher-structural interface family* is a declared class of derived, sheaf-theoretic, Fukaya-categorical, A_∞ , dg-categorical, stack-theoretic, or higher-categorical structures together with a proposed Core-readable extraction route.

Such interfaces are external enhancements unless their outputs are re-entered into the Core verifier through the protocol of this chapter.

Definition 8.61 (Search-to-proof interface family). A *Search-to-proof interface family* is a declared route by which Search artifacts, validity maps, lifted candidates, advisory indicators, or AI-agent outputs are transformed into proof objects, bridge candidates, or certificate records.

Search-to-proof interfaces do not upgrade Search artifacts into theorems without independent verifier-side proof data.

Proposition 8.62 (Interface families are admitted by protocol shape only). *The interface families listed above are admitted only at the level of protocol shape. Their detailed mathematical correctness must be proved separately in the relevant appendix, companion, bridge program, or domain-specific theorem.*

Proof. This chapter defines the general conditions for Core entry. It does not prove the correctness of any specific arithmetic, PDE, geometric, categorical, or Search-to-proof realization. Therefore each family is admitted only as a class of possible interfaces, not as a proved implementation. $\square$

8.13. Bridge programs and promotion of interface claims

Definition 8.63 (Bridge candidate). A *bridge candidate* is a proposed problem-specific implication connecting an external-domain statement and a Core-visible statement, but not yet proved as an interface transport theorem.

Definition 8.64 (Bridge program). A *bridge program* is a structured proof project whose purpose is to turn a bridge candidate into a transport theorem, or to classify why such promotion fails.

Definition 8.65 (Promotion of interface claim). An interface claim is promoted only when the following are supplied:

- (i) a precise statement;
- (ii) declared hypotheses;
- (iii) declared source and target layers;
- (iv) a purpose-faithful realization route;
- (v) all required constructibility conditions;
- (vi) all required eligibility conditions;
- (vii) all required tower comparison artifacts;
- (viii) all required ledgered defects;

- (ix) artifact references;
- (x) a proof sufficient for independent reconstruction;
- (xi) an explicit status change from $[Spec]$, candidate, advisory, or Search artifact to theorem-level status.

Proposition 8.66 (No silent promotion). *No interface claim, bridge candidate, advisory signal, validity map, Search report, or AI-generated explanation is promoted to theorem status silently.*

Proof. The Claim Taxonomy requires explicit status. The Core closure principle forbids enlargement by Search, platform orchestration, or undeclared transport. Therefore promotion requires an explicit proof and status change. $\square$

8.14. Explicit non-claims of Chapter 8

The following are not asserted in this chapter.

- (i) No arithmetic theorem is proved.
- (ii) No PDE regularity theorem is proved.
- (iii) No geometric normalization theorem is proved.
- (iv) No categorical comparison theorem is proved.
- (v) No symplectic, Fukaya-categorical, derived, or higher-categorical theorem is proved.
- (vi) No external source object is automatically a Core object.
- (vii) No high-dimensional projection datum is automatically a Core certificate.
- (viii) No purpose-faithful realization hypothesis is automatically true.
- (ix) No projected object is automatically constructible.
- (x) No raw-domain observable replaces $\text{Eval}_{i,w,\tau}(F)$.
- (xi) No external Ext-group is automatically the Core group $\text{Ext}^1(\mathcal{R}(C_{\tau,w}F), k)$.
- (xii) No pre-gate replaces B-Gate⁺.
- (xiii) No advisory quantity replaces monitored Type IV diagnostics.
- (xiv) No scalar score, potential, warning flag, Search rank, or AI confidence value replaces the budget condition.
- (xv) No Core pass implies an external-domain theorem without a transport theorem.
- (xvi) No external-domain theorem follows merely because the projection appears natural, canonical, elegant, or empirically stable.

- (xvii) No case study, calibration example, computational experiment, validity map, Search report, platform artifact, or companion document enlarges the Core by appearing elsewhere in the project.

Remark 8.67 (Why these non-claims are necessary). Interface material is persuasive precisely because it is concrete. For that reason, the protocol must explicitly refuse the inference

concrete realization $\implies$ theorem-bearing conclusion about the external problem.

Without that refusal, the boundary fixed in Chapter 7 would immediately be eroded.

8.15. Main-body closure

Theorem 8.68 (Interface protocol package). *The main body closes with the following interface protocol package.*

- (i) *External objects may be connected to the AK Core only through declared projection or realization data.*
- (ii) *High-dimensional projection is the Interface-side mechanism that creates Core-readable inputs.*
- (iii) *Collapse-based degeneration judgment is the Core verifier-side mechanism applied to repaired windowed collapsed persistence data.*
- (iv) *The projected object must have a declared Core-readable target.*
- (v) *Constructibility is mandatory for persistence-level Core use.*
- (vi) *The eligible regime is mandatory whenever the Ext^1 -clause is invoked.*
- (vii) *All Core-relevant comparisons are evaluated through*

$$\text{Eval}_{i,w,\tau}(F) = T_{\tau}^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)).$$

- (viii) *Problem-specific implications are $[Spec]$ by default.*
- (ix) *Pre-gates may prepare domain objects, but they never replace or weaken Core gates.*
- (x) *Advisory quantities may guide inspection, but they do not determine Core verdicts.*
- (xi) *Interface certificate records must satisfy the Minimal Manifest discipline.*
- (xii) *Interface pass, Core pass, and external-domain theorem are distinct.*
- (xiii) *No interface enlarges the Core mathematical package unless the additional content is explicitly promoted and reproved under the Core guard-rails.*

Proof. Items (i)–(iii) follow from Definitions 8.5, 8.7, and 8.8. Items (iv)–(vii) follow from the declared target, constructibility, eligibility, and repaired after-collapse interface disciplines. Items (viii)–(x) follow from the default $[Spec]$ status, pre-gate/Core-gate separation, and advisory non-verdict rule. Item (xi) follows from the manifest requirement. Item (xii) follows from the definitions of interface pass, Core pass, and external-domain conclusion. Item (xiii) follows from the Core closure principle and the non-claim discipline. $\square$

Remark 8.69 (Final closure of the main body). The repaired AK Core therefore closes in the following form:

$$\begin{aligned}
X &\longmapsto \mathfrak{R}_{AK}(X) \\
&\longmapsto \text{Core-readable persistence profile} \\
&\longmapsto \text{Eval}_{i,W,\tau} \\
&\longmapsto B\text{-Gate}^+ \\
&\longmapsto \text{audited Core verdict.}
\end{aligned}$$

Any claim beyond the audited Core verdict requires a separately declared transport theorem.

This is the final main-body placement of the original AK principle: high-dimensional projection supplies the readable shadow; the Core decides only the repaired collapse behavior of that shadow; external theorems require transport.

8.16. Provenance and status

This chapter is intentionally schematic. It contains no calibration examples, no case studies, no computational experiments, no Search outcomes, no validity maps, and no problem-family-specific theorems. Its role is to make later appendical, companion, platform, and interface material readable without allowing that material to upgrade the theorem-bearing status of the Core.

The material in this chapter depends on:

- (i) the Claim Taxonomy and Minimal Manifest discipline of Chapter 1;
- (ii) the repaired barcode-level collapse and windowed evaluation policy of Chapter 2;
- (iii) the eligible realization and one-way bridge discipline of Chapter 3;
- (iv) the monitored Type IV diagnostics and tower comparison artifact discipline of Chapter 4;
- (v) the certificate-record and gluing discipline of Chapter 5;
- (vi) the Core sovereignty and audit discipline of Chapter 6;
- (vii) the non-claim and Core closure principles of Chapter 7.

Status labels for this chapter. *Core / interface definitions:* Definitions [8.4](#), [8.5](#), [8.7](#), [8.8](#), [8.11](#), [8.13](#), [8.15](#), [8.16](#), [8.19](#), [8.20](#), [8.23](#), [8.27](#), [8.30](#), [8.31](#), [8.32](#), [8.35](#), [8.36](#), [8.37](#), [8.38](#), [8.39](#), [8.43](#), [8.44](#), [8.45](#), [8.48](#), [8.49](#), [8.52](#), [8.53](#), [8.54](#), [8.57](#), [8.58](#), [8.59](#), [8.60](#), [8.61](#), [8.63](#), [8.64](#), [8.65](#).

Core / interface propositions and theorems: Propositions [8.9](#), [8.17](#), [8.21](#), [8.25](#), [8.29](#), [8.33](#), [8.46](#), [8.50](#), [8.55](#), [8.62](#), [8.66](#), Theorem [8.41](#), Theorem [8.68](#).

Interpretive doctrine: Remarks [8.1](#), [8.2](#), [8.3](#), [8.6](#), [8.10](#), [8.12](#), [8.14](#), [8.18](#), [8.22](#), [8.24](#), [8.26](#), [8.28](#), [8.34](#), [8.40](#), [8.42](#), [8.47](#), [8.51](#), [8.56](#), [8.67](#), [8.69](#).

Explicit exclusions: Section [8.14](#).

Main-body closure: Theorem [8.68](#).